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AI-Based Synthetic Users for Early-Stage UI/UX Validation

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The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

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Abstract

User Experience (UX) validation is an important step in digital product design, yet traditional human-led testing is often too resource-heavy to conduct during the early conceptual design phases. While "Shift-Left" economics heavily prefer early validation, existing automated solutions either rely on coded environments or utilize fully autonomous AI agents that generate machine-readable, unexplainable feedback, leading to a lack of designer trust. This thesis addressed these limitations by exploring how Generative Artificial Intelligence (GenAI) could be utilized to conduct early-stage usability testing. The aim was to design and evaluate an AI-based synthetic user framework capable of evaluating static UI/UX prototypes while prioritizing behavioral realism and designer trust over total autonomous scale. Following a Design Science Research (DSR) methodology, the artifact was iteratively developed over three prototype phases. Essentially, this prototype was developed not as a means in itself but as a means to an end; i.e., it served as an exploratory framework to determine the extent to which GenAI can reliably execute UX validation. To prevent the Large Language Model (LLM) from defaulting to a generic technical assistant, the framework utilized parameter-free optimization. Specifically, advanced prompt engineering architectures (Intent, Domain, and Demand constraints combined with Chain-of-Thought) were combined with the 15-facet BFI-2 psychometric framework to force the AI to simulate subjective human traits and output actionable cognitive friction. Evaluation with a single domain expert, who also collaborated in the framework's design, suggested that while synthetic users can successfully identify valid usability issues and build designer trust through transparent reasoning, they still show an inherent "rationality bias" and cannot fully replicate human unpredictability. The findings concluded that GenAI shows high potential as an early-stage, human-in-the-loop decision-support tool, but must complement rather than replace real human testing.

Keywords: Synthetic users; UI/UX validation; Generative AI; Human-in-the-loop; Design Science Research

Sammanfattning

Validering av användarupplevelse (UX) är ett viktigt steg vid design av digitala produkter, men traditionella, människodrivna användartester är ofta för resurskrävande för att genomföras i de tidiga konceptuella faserna. Även om "Shift-Left"-principer starkt förespråkar tidig validering, förlitar sig nuvarande automatiserade lösningar antingen på kodade miljöer eller på helt autonoma AI-agenter som genererar maskinläsbar och oförklarlig feedback, vilket leder till bristande tillit hos designers. Denna studie adresserade dessa begränsningar genom att utforska hur generativ artificiell intelligens (GenAI) kan användas för att genomföra användartester i tidiga skeden. Syftet var att designa och utvärdera ett AI-baserat ramverk med syntetiska användare som kan utvärdera statiska UI/UX-prototyper, där beteendemässig realism och användartillit prioriteras framför fullständig autonomi.

Genom att tillämpa Design Science Research (DSR) som metodologi utvecklades artefakten iterativt under tre prototypfaser. I grund och botten utvecklades inte denna prototyp som ett självändamål, utan snarare som ett medel för att nå ett mål; den fungerade som ett utforskande ramverk för att fastställa i vilken utsträckning GenAI pålitligt kan utföra UX-validering. För att förhindra att språkmodellen (LLM) agerar som en generisk teknisk assistent använde ramverket parameterfri optimering (parameter-free optimization). Specifikt integrerades avancerade "prompt engineering"-arkitekturer (Intent, Domain och Demand i kombination med Chain-of-Thought) med det psykometriska BFI-2-ramverket (15 fasetter) för att tvinga AI:n att simulera subjektiva mänskliga egenskaper och generera agerbar kognitiv friktion.

Utvärderingen, som genomfördes tillsammans med en enskild domänexpert, som även var delaktig i utvecklingen av verktyget, antydde att även om syntetiska användare framgångsrikt identifierar relevanta användbarhetsproblem och bygger upp tillit genom transparenta resonemang, uppvisar de fortfarande "rationalitetsbias" (rationality bias) och kan inte fullt ut återskapa mänsklig oförutsägbarhet. Sammanfattningsvis drog studien slutsatsen att GenAI visar stor potential som ett beslutsstöd (human-in-the-loop) i tidiga skeden, men att det måste komplettera, snarare än ersätta, tester med riktiga människor.

Nyckelord: Syntetiska användare; UI/UX-validering; Generativ AI; Human-in-the-loop; Designvetenskaplig forskning

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Chapter 1

Introduction

1.1 Introduction

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User validation testing [1] is a crucial part of Human-Centered Design (HCD), providing the empirical evidence required to help align digital interfaces with human behaviors. By measuring how users perceive and interact with an interface during specific tasks, designers can identify and mitigate usability risks long before the actual product reaches its end destination. In the more modern software development life-cycle, the focus has shifted more towards "Shift-Left" testing [4], where validation occurs as early as possible to avoid the increased cost of fixing design flaws during the development phase. This raises the question of how early-stage UI/UX validation can be supported before implementation, particularly for static design artifacts such as Figma prototypes, and how GenAI-based synthetic users can be designed and evaluated as a human-in-the-loop decision-support tool. Investigating this is in scope of the thesis at hands.

1.2 Background

1.2.1 Context

Established methods for user validation like the ones described in "the sprint book" [15], focus on observation and interviewing of real users during predetermined tasks. These methods give the developers strong qualitative insights, but at the high cost of human testers. Sourcing user testers is often a challenge in itself, even without the cost in mind, especially in smaller cities. This thesis was conducted in collaboration with Strawberry Planet, a company specialized in digital product development. Strawberry planet uses the application Figma for early stage UI/UX-design and validation.

1.2.1.1 The economics of Shift-Left

The economic benefit of increasing the availability of pre-development validation is rooted in "Shift-Left" software engineering principles. As demonstrated by Andriadi et al. [4], moving testing to the earliest stages of an agile development cycle significantly reduces production bugs and improves overall software quality. Despite these

[1] User validation testing is defined from a UI/UX designers perspective, see Section 1.2.2

documented benefits, a recent review by Alenezi and Akour [2] highlighted that while AI is heavily utilized for specialized tasks like code generation, it remains severely underutilized in early-stage conceptual design. Traditional Shift-Left methodologies primarily rely on evaluating coded artifacts in lower development environments.

1.2.1.2 Prompt Engineering

To help address the unreliability of standard LLMs in specialized tasks, recent research has formalized prompt engineering into structured taxonomies [12,16]. Specifically, Liu et al. [16] propose a framework that categorizes prompt interventions into distinct structural components, mainly focusing on intent (the goal), Domain (the persona's knowledge base), and Demand (specific constraints). This structured approach demonstrates that clearly identifying the domain and demand can help prevent the LLM from giving generic answers as well as mitigating "content bias", which is the tendency of an AI to revert to a standard, helpful assistant rather than maintaining a specific persona.

Furthermore, achieving human-like problem-solving requires specific cognitive framing. Both foundational prompt engineering principles and recent taxonomic frameworks [12, 16] highlight that techniques such as Chain-of-Thought (COT) and Plan-Execute Decomposition are very important to enable models to emulate human logic. By forcing the LLM to think "step-by-step" and break down complex navigation tasks into simpler sub-goals, the model's final actions become heavily grounded in its assigned persona logic rather than its underlying algorithmic training. Additionally ,using "internal feedback" where the AI reviews its own previous steps, significantly improves the accuracy and consistency of its final output [16].

The system architecture presented in this thesis heavily leverages this taxonomy. To prevent instruction drift, the prompt formulation is divided into Intent, Domain, and Demand constraints. To enforce the Chain-of-Thought behavior necessary for a realistic Think-Aloud protocol, the system forces the LLM to generate a step-by-step reasoning trace prior to executing any action.

1.2.1.3 Agent Optimization and Persona Consistency

As LLM-based agents become increasingly utilized, optimizing their behavioral performance has become even more critical. In a comprehensive survey, Du et al. [8] categorize agent optimization into two different methods, parameter-driven methods (e.g., fine-tuning) and parameter-free methods (e.g., prompt design), noting that "standard" LLMs often fail in complex, specialized environments. The limitations of standard LLMs in UX were further confirmed by Bakaev et al. [5], who found that unconstrained models struggle to maintain a consistent simulated persona. To address this problem, prior research has demonstrated that techniques such as few-shot in-context learning and negative constraints significantly improve an LLMs ability to maintain a simulated identity.

This thesis makes direct use of the parameter-free optimization strategies identified by Du et al. [8] to solve the persona consistency problem. To prevent the LLM from acting like a highly detailed, helpful technical assistant, this work introduces a structured framework utilizing a rigid prompt structure. Specifically, the agent's

core instructions are mapped to the expanded 15-facet BFI-2 psychometric framework developed by Soto and John [20]. By employing strict negative constraints alongside these validated human personality traits (e.g., specific variations in negative emotionality or open-mindedness), our parameter-free approach forces the model to suppress its inherent bias and reliably simulate authentic, flawed human usability friction.

1.2.2 Key terminology

In this thesis, the term *user validation testing* refers to structured tests where users attempt to solve predefined tasks in a UI under observation. *Generative AI (GenAI)* refers to a category of artificial intelligence capable of generating new content or reasoning, which in this study is utilized to simulate user interactions. These simulations are powered by *Large Language Models (LLMs)*, which are advanced deep learning models trained on big datasets to process and generate human-like text and logic. An *AI-based user agent* or *synthetic user* is an LLM-driven entity designed to navigate interfaces and mimic human reasoning. This behavior is then guided by an *AI persona*, a structured description of a specific user profile (e.g., goals, frustrations, and technical knowledge) used to ensure *behavior realism*, which is the degree to which the AI's actions and reasoning match those of a real human participant. *Parameter-free optimization* is a method for guiding and adjusting the behavior of an LLM solely through structured prompt engineering and constraints. *BFI-2 (Big Five Inventory-2)* is a scientifically researched and validated psychological assessment framework that measures human personality across five major areas: Open-Mindedness, Conscientiousness, Extraversion, Agreeableness, and Negative Emotionality. A *black box* refers to a system whose internal decision-making processes are not transparent or easily interpretable by the user. In the context of LLM-based systems, this means that while inputs and outputs can be observed, the underlying reasoning and model behavior are not directly accessible, which can affect trust and interpretability in evaluation scenarios. The *Think-Aloud protocol* is a usability evaluation method in which participants verbalize their thoughts, reasoning, and decision-making processes while performing tasks. This method is used to capture user cognition and identify usability issues. In this study, the Think-Aloud protocol was simulated by the AI-based synthetic users to generate step-by-step reasoning traces during interaction with the UI. The *Critical Incident Technique (CIT)* is a qualitative interview method used to find detailed, specific past events where a process or interaction either succeeded or failed. The study adopted a *Shift-Left* approach, which is a methodology that moves testing and validation to the earliest possible stages of the development cycle to be able to identify flaws before significant resources are committed. *Human-in-the-loop* refers to a system design approach in which human users remain actively involved in the decision-making or control process. In this study, the UI/UX designer interpreted and validated the outputs generated by the AI-based synthetic users, ensuring that the system functioned as a decision-support tool rather than a fully autonomous solution.

1.2.3 Problem description

The primary problem addressed in this thesis is that early-stage UI/UX validation is often difficult to conduct continuously in practice. In the studied industry context, external user validation is usually reserved for major features or significant workflow changes, while smaller UI iterations are often validated internally because recruiting suitable participants, preparing test material, moderating sessions, and manually reviewing recordings require considerable time and effort. This creates a practical gap where early design decisions may receive limited user-oriented validation before development resources are committed. Therefore this project addressed the main question of whether AI-based user agents can effectively conduct user validation testing directly within early-stage design environments in a way that generates relevant insights. To mimic a real-world testing phase, the approach simulated multiple different personas executing the same predefined tasks. The study focused on understanding when AI-generated feedback becomes sufficiently reliable to allow designers to move forward with confidence before any resources are committed to development.

1.3 Scope

This thesis focused on the use of AI-based synthetic users for early-stage UI/UX validation, specifically within static design environments such as Figma prototypes. The study investigated how generative AI can be applied before implementation, where traditional usability testing is typically limited. The scope is restricted to:

- Early-stage design artifacts (e.g., static screens and clickable prototypes)
- Task-based usability validation scenarios
- Simulation of user behavior through LLM-based personas
- A controlled, human-in-the-loop evaluation process

The study did not include:

- Validation of fully implemented systems (e.g., live applications or HTML environments)
- Quantitative performance benchmarking (e.g., time, cost efficiency comparisons)
- Accessibility testing requiring specialized tools (e.g., screen readers)
- Large-scale user studies with statistically generalizable results

These limitations were intentional to ensure a focused investigation of behavioral realism, insight quality, and practical usefulness of AI-generated user simulations in early design phases. The findings were therefore positioned as indicative rather than generalizable, with emphasis on practical applicability within similar UI/UX workflows.

1.4 Research Aim and Research Questions

Based on the identified problem and research gap, this thesis formulates a set of research questions to guide the investigation. The questions are structured to reflect a progression from understanding current practices, to designing a solution, and finally evaluating its effectiveness.

This structure follows the principles of Design Science Research, where problem identification, artifact development, and evaluation are conducted in a sequential and iterative manner. By organizing the research in this way, the study ensures that the proposed solution is grounded in real-world needs and systematically assessed in terms of its practical usefulness.

The following subsections present the overall aim of the study and the research questions that guide the investigation.

1.4.1 Aim

This thesis explores to which extent AI can facilitate early-stage UI/UX validation within a digital industry development environment. By incorporating the "Shift-Left" approach, moving the testing to the earliest phase, this study investigated whether an AI-based framework could provide trustworthy and actionable usability insights to support UI/UX designers during early-stage development. To ensure that the study was based on real-world problems, we followed a requirements-driven approach that will be divided into three distinct phases: (1) an information gathering phase to map current manual validation processes and identify automation potential, (2) an iterative development phase focused on addressing specific designer challenges, and (3) a final evaluative phase. To this end, we formulated three research questions that map to these stages of the investigation:

1.4.2 Research Questions

Table 1.1: Research Questions and Methodological Mapping

ID	Research Question	Method / Data Source
RQ1	How is early-stage UI/UX validation currently conducted in practice, and which activities have the greatest potential for GenAI-based automation?	Semi-structured interview; workflow analysis, thematic analysis
RQ2	How can an AI-based synthetic user framework utilizing personas be designed and refined to effectively support early-stage UI/UX validation?	Constructive research
RQ3	How is the AI-based synthetic user framework perceived by a UI/UX expert in terms of usefulness, behavioral realism, trust, and practical applicability for early-stage UI/UX validation?	Qualitative Case study, Expert Evaluation form

1.5 Outline

The remainder of this thesis is structured as follows.

Chapter 2 presents the related work, reviewing existing research on automated usability evaluation, synthetic users, and AI-driven UX validation, and identifies the research gap addressed in this study.

Chapter 3 describes the research methodology, including the Design Science Research approach, data collection methods, and the design and development of the proposed framework.

Chapter 4 presents the results and analysis, including findings from the empirical study, the developed prototypes, and the evaluation of the framework.

Chapter 5 discusses the findings in relation to the research questions, highlighting key insights, implications, and limitations of the study.

Chapter 6 concludes the thesis by summarizing the main contributions, answering the research questions, and outlining directions for future work.

Chapter 2

Related Work

2.1 Related Work

2.1.1 Introduction

The improvement of User Experience (UX) validations is a crucial step spanning classical human-computer interaction (HCI) and modern artificial intelligence. This chapter reviews the scientific literature directly related to usability testing using LLMs, tracing its evolution from rigid, legacy algorithmic methods to modern LLM-driven synthetic agents. By examining parallel innovations in agent-driven evaluation, modern synthetic UX testing frameworks, and the critical need for Human-AI collaboration, this chapter identifies the exact research gaps addressed by this thesis. This review demonstrates the increasing need for high-fidelity, human-in-the-loop personas that serve as early-stage decision support rather than fully autonomous, robotic evaluators.

2.1.2 Legacy Methods in Usability Automation

Historically, earlier attempts to automate usability evaluation were confined to rigid, algorithmic assessments rather than human behavioral simulation. In their foundational survey on automated usability evaluation, Ivory and Hearst [14] categorized the current state of the art based on the role of automation, with a heavy emphasis on analytical modeling and simulation. However, these early systems required immense manual effort to program and were generally limited to predicting error-free, expert performance. Most importantly, the authors explicitly acknowledged that these automated methods were intended to complement human testing, as they could not capture subjective user preferences, emotions, or satisfaction. Even as legacy methods improved, their focus remained very mechanical. For example, Cassino et al. [6] successfully validated an automated GUI evaluation method that decoupled testing from backend dependencies, functioning entirely on the interface level. A very critical constraint of this approach, however, was its strict confinement to assessing structural "consistency aspects". This thesis moves away from this legacy model. By replacing deterministic rule-checking with structurally constrained LLM personas, our approach moves the automation past basic visual consistency and into more high-fidelity, subjective behavioral simulation.

2.1.3 Parallel Innovations in Agent-Driven Evaluation

The use of generative multi-agent systems has shown good success in domains requiring complex behavioral simulation. In the social sciences, Park et al. [19] demonstrated that LLMs could flexibly simulate believable human routines, beliefs, and social dynamics without the need for hand-coded rules. Similarly, in the mechanical engineering world, Chen et al. [7] developed DesignAgent, an LLM-based framework that automatically evaluates the structural feasibility of early-stage physical product concepts. While these works prove the broader viability of agent-driven evaluation in both social sciences and physical engineering, the potential for specializing these architectures specifically for digital UI/UX validation within the software design lifecycle remains largely unexplored.

2.1.4 Modern Synthetic UX Testing

Applications of LLM agents to UX testing have recently emerged to address the high resource costs of traditional human testing. As highlighted in a systematic review by Vukorepa [21], "Synthetic Users" offer a time-efficient way to generate usability data. Technologically, Lu et al [17] introduced UXAgent, which demonstrated the feasibility of generating usability logs via LLMs. Building upon this, Holter et al. [13] proposed UXCascade, which provides multi-level analysis to help designers interpret vast amounts of AI-generated UX logs.

Despite the technical success, significant limitations remain regarding behavioral fidelity. By attempting to fully automate the testing process end-to-end, Lu et al. [17] found that human reviewers perceived the agents reasoning as "too detailed" and highly robotic. Furthermore, both UXAgent and UXCascade rely heavily on interacting with HTML-based environments, which delays validation. To bring evaluation earlier in the design cycle, recent work by Duan et al. [9] integrated GPT-4 directly into Figma to automate heuristic evaluations on static mockups. While this demonstrates the value of a human-in-the-loop approach, allowing designers to iteratively accept or dismiss AI suggestions, the tool simulates an expert evaluator strictly applying design guidelines rather than a synthetic user navigating a task. Therefore, a gap remains between early-stage mockup evaluation and synthetic user testing. Current autonomous systems often sacrifice behavioral fidelity for automation. To produce actionable and realistic usability friction, future approaches must investigate how to integrate human-like cognitive methodologies, such as the Think-Aloud protocol [15, 18] without relying on fully autonomous, end-to-end execution.

2.1.5 Human-AI Collaboration and Designer Trust

Integrating generative models into design workflows presents some unique collaborative challenges. In a recent systematic literature review of LLMs in UI/UX design, Ahmed and Imran [1] identified severe ongoing limitations, specifically citing prompt instability and limited explainability as barriers to adoption. Consequently, the review emphasized the need for Human-AI collaboration over total automation. Furthermore, achieving successful collaboration relies heavily on the designer's trust in the AI's feedback. Exploring this further, Yoon et al. [24] investigated designer-AI

collaboration during early-stage design, highlighting the critical need for AI tools to provide transparent, trustworthy recommendations. Addressing these trust and explainability challenges require a shift away from highly autonomous, black-box evaluators. As identified by Ahmed and Imran [1] and Yoon et al. [24], there is a huge need for collaborative, human-in-the-loop systems that prioritize qualitative, human-like reasoning over robotic, algorithmic logs to successfully promote transparency and designer trust.

2.1.6 Summary and Research Gap

The literature reviewed in this chapter demonstrates a clear progression in usability validation, going from rigid algorithmic tools to modern generative agents. However, a significant research gap remains at the intersection of early-stage validation, behavioral fidelity, and designer trust. While previous works have successfully explored automated usability evaluation, legacy systems lack human behavioral fidelity. At the same time, modern synthetic frameworks suffer from robotic persona generation and leaves out human designer a lot due to their reliance on total automation. Specifically, no prior work was found that successfully combines early-stage static validation, persona-driven simulation, and strict human-in-the-loop control. This thesis directly fills these gaps by leaving the pursuit of a fully autonomous software pipeline. Instead, it processes sequential, static screenshots and employs a collaborative Human-AI workflow where the human designer orchestrates the state changes. By positioning the AI strictly as an explainable decision-support tool, this approach ensures that the synthetic personas prioritize authentic behavioral friction and designer trust over autonomous scale. Table 2.1 summarizes the core differences between the established and the approach proposed in this thesis.

Table 2.1: Comparison of Related Works and the Addressed Research Gaps

Author(s)	**Domain Focus**	**Key Limitations in the Literature**	**How this Thesis Fills the Gap**
Ivory & Hearst [14]	Usability Evaluation Survey	Acknowledged that automated methods fundamentally fail to capture subjective user preferences or emotions.	Transitions automation from rigid, objective metrics to subjective, cognitive simulation.
Cassino et al. [6]	Legacy UI Automation	Built an evaluation method strictly confined to assessing mechanical visual "consistency aspects".	Replaces deterministic mathematical checks with LLM-driven behavioral friction.

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Author(s)	Domain Focus	Key Limitations in the Literature	How this Thesis Fills the Gap
Chen et al. [7]	Generative Engineering	Tailored agents for physical product feasibility, leaving digital UX validation unaddressed.	Shifts the agent-driven evaluation model into the digital UI/UX lifecycle.
Park et al. [19]	Generative Social Agents	Proved agents can simulate human routines, but the sandbox environment lacks digital UI constraints.	Adapts generalized agent role-play specifically for targeted software design validation.
Lu et al. [17]	Synthetic UX Automation	UXAgent prioritized total automation, resulting in robotic, overly detailed feedback lacking human realism.	Abandons total automation for a Human-in-the-Loop approach to ensure authentic cognitive friction.
Holter et al. [13]	UX Data Analysis	UXCascade relies heavily on functional HTML environments, delaying validation.	Shifts validation earlier by processing sequential, static screenshots.
Duan et al. [9]	Automated Heuristic Evaluation	Integrated LLMs into design tools to simulate an expert evaluator strictly applying predefined design guidelines.	Moves beyond expert heuristic evaluation by simulating authentic end-user cognitive friction via the Think-Aloud protocol.
Yoon et al. [24]	Early-Stage AI Collaboration	Designers struggle to trust generative AI tools during early ideation without transparent verification.	Integrates explicit collaborative oversight, ensuring designers can verify AI reasoning step-by-step.
Ahmed and Imran [1]	Human-AI Trust	Designers experience "algorithmic aversion" toward highly autonomous, unexplainable AI outputs.	Positions the AI strictly as an explainable decision-support tool rather than an autonomous evaluator.

Chapter 3

Method

3.1 Method

The overarching methodology for this thesis was Design Science Research (DSR) [22]. This methodology was chosen because the main objective of the research was to design, implement, and iteratively evaluate an artifact, an AI-based synthetic user framework to address a real-world industry challenge. DSR provides the necessary structured framework for developing and evaluating such artifacts to solve identified organizational problems.

which provides the framework for developing and evaluating innovative artifacts to solve identified organizational problems. Under the Design Science Research umbrella, the research was executed as an evaluative case study at Strawberry Planet. This case study served as the environment for the DSR design cycle, beginning with an exploratory RQ1 for problem identification and automation potential. The study was focused on developing an AI-based framework focused on testing UI designs during their early phase development and evaluating its practical performance against professional standards. The thesis was conducted in three phases, first a problem identification phase then a research and development phase both using qualitative methods (RQ1-RQ2); finally, an evaluation and validation phase (RQ3) that concluded the investigation. The research design can be seen in table 3.1 following the decision making structure of *Experimentation in Software Engineering* [23]

Research design element	**This thesis**
Research outcome	Applied research
Research logic	Mixed: Inductive (RQ1, RQ2) + Deductive (RQ3)
Research purpose	Exploratory (RQ1), Evaluative (RQ2, RQ3)
Research paradigm	Mixed: Interpretivist (RQ1, RQ2), Positivist (RQ3)
Research process	Mixed methods
Research methodology	Design Science research
Data Collection	Semi-structured interviews, iterative prototyping, expert evaluation
Data analysis	Thematic analysis + Descriptive analysis

Table 3.1: Overall research design positioning according to the decision-making structure [23].

3.1.1 Problem identification RQ1

RQ1 was answered through a semi-structured interview utilizing critical incident technique (CIT) [10] to identify current problems, limitations and practical challenges in early stage UI/UX validation. The interview aimed to identify the current validation workflow, what activities need human-in-the-loop involvement and what could be automated through GenAI.

The Critical Incident Technique was used to structure parts of the interviews around concrete examples of previous validation experiences rather than only general opinions. This was suitable for RQ1 because the aim was to identify practical problems, successful practices, and automation opportunities in real early-stage UI/UX validation work.

The following list presents the main interview questions used to guide the interviews. For the full interview guide, including sub-questions and detailed activity structure, see Appendix A.2.

- Can you describe your current role and responsibilities at [organization]?
- How long have you been working with the current Figma-based design cycle?
- On a typical project, how often do you perform manual usability or "sanity" checks on prototypes?
- Can you describe the last time early-stage UI validation was frustrating or inefficient?
- Can you describe a validation session that went very smoothly?

- Walk me through the most recent validation session you conducted.
- What are the most time-consuming aspects of traditional user testing for you?
- Which usability flaws are often missed during the Figma phase and only found in development?
- What would make an AI-generated "think-aloud" log trust-worthy enough for you to act on?
- Where do you see the most significant potential for GenAI-based automation in your current workflow?
- If you were to use automation tools, what aspect of it would you see the most gain from? Speed Vs Accuracy
- Can you walk me through how you currently evaluate your design, what do they usually look like during the early validation stage?

Prior to the interviews, a researcher familiarization phase was conducted to establish domain understanding and reduce interpretation bias. The interview guide consisted of six structured activities, including introductory questions, CIT-based incident elicitation (positive, negative, and neutral), and additional questions targeting automation potential, workflow bottlenecks, and trust in AI-based systems. The full guide is provided in Appendix A.2

The target population consisted of UI/UX designers and product developers who are involved in early-stage validation and usability testing. Participants were selected using purposive sampling based on their experience in UI/UX validation and involvement in early-stage design processes. To start with, only the primary UI/UX expert at Strawberry Planet was interviewed and acted as a key informant, with focus on single organizational context to ensure artifact–workflow alignment and practical applicability. Two other UI/UX designers and one requirements engineer that work closely with UI/UX designers were also interviewed to get organizational breadth and variations in the workflow. In total, there were four interviews conducted for RQ1.

The collected qualitative data was then analyzed using workflow analysis combined with thematic analysis to identify recurring patterns, workflow, tools & practices, challenges & pain points, bottlenecks, automatization candidates and human-in-the-loop boundaries. The analysis followed an inductive thematic approach, where interview data was coded and grouped into recurring themes seen in Table 4.1 in Results of RQ1, section 4.1. The thematic analysis was conducted collaboratively by the authors rather than through independent parallel coding. Codes and themes were discussed and refined jointly based on the interview material and the focus of RQ1. No formal inter-coder agreement metric was calculated, and no formal member checking was conducted.

The results of RQ1 directly influence the design decisions in RQ2 and the evaluation metrics in RQ3. RQ1 contributed to defining a Minimum Viable Product (MVP) for Strawberry Planet and clarifying the functional scope and priorities of the AI-based framework.

3.1.2 Design & Development RQ2

RQ2 focused on the design and refinement of the AI-based synthetic user framework with emphasis on behavioral realism and practical usefulness. This phase constituted the design and development phase of the overarching Design Science Research methodology, rather than a separate methodological approach, and was conducted using a qualitative, interpretive approach. The purpose was evaluative, aiming to iteratively improve the artifact through continuous expert feedback. Accordingly, the development process followed an iterative build–evaluate–refine cycle, where design decisions were continuously assessed and adjusted based on observed output behavior and expert feedback.

Prior to the main RQ2 development phase, an initial scoping and design exploration was conducted to refine the design space and define a feasible direction for the artifact.

3.1.2.1 Technical Feasibility Assessment

Prior to the main design and development phase, a brief technical feasibility assessment was conducted to verify that current multi-modal LLMs are capable of interpreting UI screenshots, which is a prerequisite for the proposed framework. The assessment explored the ability of different models to identify and reason about interface elements. The results of this assessment are presented in Section 4.2.1 and informed the decision to proceed with the selected design approach.

3.1.2.2 Design Exploration and Alternative Identification

During an initial stage of development, it became evident that multiple architectural approaches could be used to implement the synthetic user framework. This introduced challenges in defining scope and feasibility within the constraints of the thesis.

To address this, an exploratory design phase was conducted where different design directions were explored and assessed by both usefulness for the UI/UX designer and feasibility from a technical stand point. The purpose of this phase was to investigate different design directions and better understand trade-offs between key dimensions such as automation, controllability, reliability, and behavioral realism.

The exploratory design phase resulted in three conceptual design options, described in Section 4.2.2.2 The three designs was then compared across eight dimensions, automaton level, reliability, setup effort, designer effort, scalability, output quality, technical risk, and whether it works early in the design phase. See the comparison in Table 4.2

This phase also highlighted that increasing the level of automation introduced significant technical challenges.

3.1.2.3 Evaluation and Selection of Design Direction

Following the exploratory phase, the identified design directions were discussed and evaluated together with the UI/UX expert. The eight dimensions mentioned earlier were used as a basis for comparing the three design options.

During the selection of a Minimum Viable Product (MVP), input from the UI/UX expert introduced an important decision criterion. It was emphasized that the value of automation is fundamentally dependent on the quality of the generated outputs, and that increased automation would not be beneficial if the resulting insights lack realism or usefulness.

This observation led to a refinement of the research focus. Instead of prioritizing a highly automated solution, the development was deliberately scoped to focus on the quality, realism, and usefulness of LLM-generated outputs. This shift allowed the study to more effectively investigate whether synthetic users can produce meaningful reasoning and actionable usability insights.

Based on these considerations, a controlled and less automated approach was selected as the MVP, enabling greater transparency, reproducibility, and evaluation of the AI-generated behavior.

The subsequent RQ2 development phase followed an iterative build–evaluate–refine cycle, in which multiple prototype versions were developed and evaluated to analyze how specific issues in the generated outputs emerged, evolved, and were mitigated across design iterations. The final prototype resulting from this iterative process was subsequently evaluated in RQ3.

3.1.2.4 Development Approach

The selected approach was implemented as an iterative build-evaluate-refine cycle, consistent with the Design Science Research principles. The framework operates through a controlled interaction loop, where the AI-based synthetic user processes one UI state at a time and generates corresponding reasoning and suggested actions before progressing. The actions are then interpreted manually by the UI/UX designer and then respond with the next correlating image.

This design ensures that the AI does not have access to the full navigation structure simultaneously, reducing bias and improving the realism of simulated interaction.

The same UI designs and underlying task goals were maintained across prototypes to ensure consistency in the evaluation context and enable meaningful comparison between iterations. However, personas and task prompt formulations were iteratively refined across prototypes to improve behavioral realism and align generated outputs with expected user behavior.

It also allows the interaction flow to remain verifiable and reproducible, which is essential for evaluating the quality of the generated outputs.

The framework generates outputs such as think-aloud reasoning logs and structured interaction steps, simulating how a user would approach and solve predefined tasks within a UI.

3.1.2.5 Expert Involvement and Iterative Refinement

Throughout the development process the UI/UX expert from Strawberry Planet participated both as a domain informant and a formative evaluator of the framework. This involvement ensured that the design and refinement of the artifact remained grounded in real-world UX practices and aligned with professional evaluation standards.

Feedback sessions were conducted iteratively as part of the build–evaluate–refine cycle. During these sessions, intermediate outputs generated by the framework were reviewed and discussed with the expert. The evaluations focused on key aspects relevant to early-stage UI/UX validation, including behavioral realism, insight quality and practical usefulness.

This qualitative feedback guided successive design decisions, particularly in refining persona definitions, prompt structures, and interaction flow. The iterative nature of this process allowed identified issues in earlier prototypes to be addressed and improved in subsequent versions of the framework.

Through these iterative evaluation sessions, recurring patterns were identified in the generated outputs regarding:

- **Behavioral realism**, such as differences between natural and overly rational behavior
- **Insight quality**, including the relevance and clarity of identified usability issues
- **Actionability**, referring to whether the outputs supported concrete design decisions
- **Usefulness**, in terms of overall value within a UI/UX workflow
- **Unrealistic or misleading reasoning**, including incorrect assumptions or overconfident conclusions

Each identified pattern was interpreted in relation to these criteria to assess how well the framework supported realistic and meaningful usability evaluation. Rather than applying formal coding schemes, the analysis followed a qualitative and descriptive approach based on structured expert judgment.

The expert assessed the extent to which the simulated interactions reflected realistic user actions, reasoning processes, and common usability issues based on prior experience with human user testing. This form of evaluation is consistent with established practices in UX, where expert interpretation plays a central role in assessing usability and user behavior.

The continuous involvement of the expert enabled iterative refinement of the framework, contributing to improved behavioral realism, higher-quality insights, and better alignment with practical UX validation workflows.

3.1.2.6 Technical Scope

The technical scope of RQ2 was limited to screenshot-based UI interpretation, predefined task execution, persona-based simulation, and generation of interaction traces for expert review. These scope boundaries where used to keep the artifact feasible within the thesis timeframe and to support comparison across prototype iterations. Test scenarios were designed in collaboration with Strawberry Planet, using realistic validation tasks derived from actual design workflows. Multiple simulation runs were conducted across different personas to explore variation in behavior and reasoning.

The concrete framework design and resulting outputs are presented in Chapter 4

3.1.2.7 Multi-model evaluation of synthetic users

To reduce model-specific bias and improve the robustness of the evaluation, the synthetic personas were tested across multiple LLMs. Based on earlier feasibility exploration in Section 4.2.1, models with insufficient performance were excluded, and the final evaluation focused on the most suitable candidates.

After the development of the final prototype, the same personas, designs, and task scenarios were executed across the selected LLMs.

This allowed comparison of:

- Behavioral realism across models
- Consistency in reasoning and interaction patterns
- variation in generated insights

By holding personas, prompts, and tasks constant across models, differences in output can be attributed to the underlying LLM rather than the experimental setup.

3.1.3 Evaluation RQ3

The purpose of RQ3 was to evaluate how the final AI-based synthetic user framework was perceived by the UI/UX expert, in relation to early-stage UI/UX validation. While RQ2 focused on the design and refinement of the framework, RQ3 focused on its perceived usefulness, practical applicability, and remaining limitations from the perspective of a UI/UX expert.

The evaluation was conducted as a qualitative expert evaluation session within the context of Strawberry Planet. One practitioner applied the framework to a realistic UI/UX task and subsequently reflected on its usefulness through an evaluation form. The goal was not to measure the framework statistically or compare it quantitatively against human user testing, but to explore whether the framework could provide value as a decision-support tool in a realistic UI/UX workflow.

3.1.3.1 Evaluation approach

The evaluation was designed as a final summative evaluation of the completed framework. The UI/UX expert from Strawberry Planet participated in a structured demonstration and evaluation session. The expert had prior experience with usability validation and was familiar with the design context used in the study.

The evaluation focused on perceived usefulness rather than technical performance. This decision was based on the development process in RQ2, where it became clear that the main challenge was not whether the framework could generate outputs, but whether these outputs were useful, trustworthy, and actionable enough for a practitioner to consider using them in practice.

3.1.3.2 Evaluation Design

The selection of the evaluated models were based on their ranking as the top-performing Large Language Models (LLMs) on current industry leaderboards at the

time of this study. As mentioned previously in the technical feasibility assessment (Section 4.2.1), OpenAI (GPT-5.4) , Claude (Claude Opus 4.6), and Gemini (Gemini 3.1 Pro-Preview) were initially tested on their ability to accurately interpret UI screenshots. Because the framework relies entirely on the synthetic user's ability to "see" and reason about specific interface elements to perform validation tasks, accurate UI element identification is highly important. During the initial feasibility test, Gemini failed to identify these UI elements with sufficient detail compared to the other two models. Because of that, Gemini was excluded, and only OpenAI and Claude proceeded to the later stages of the study. OpenAI was chosen as the main LLM for the prototype development and testing as it was already available within the case organization's existing tools. Claude, having also passed the feasibility test, was used for the final evaluation as well to provide a comparative baseline.

Therefore, the evaluation was split into two parts. First, the UI/UX expert performed one usability validation task using the framework through OpenAI. During this session, the expert stepped through the test scenario and reviewed the generated think-aloud logs as they were produced. This allowed the expert to experience the framework as it would be used in practice, including the manual interpretation of the generated reasoning and actions.

Second, the expert was shown one completed validation run generated using Claude. This run used the same general framework structure and was included to explore whether another LLM produced outputs that appeared different or improved in terms of realism, clarity, or usefulness. The Claude run was not treated as a formal model benchmark, but as a supplementary comparison to support reflection on the framework's dependence on the underlying LLM.

3.1.3.3 Procedure

The evaluation session followed a structured procedure:

1. The researchers introduced the purpose of the evaluation and explained the framework.

2. The UI/UX expert was given a short demonstration of how the framework should be used.

3. The expert performed one usability validation task using OpenAI.

4. During the task, the expert reviewed the generated think-aloud logs and interaction reasoning.

5. After the OpenAI run, the expert reviewed one completed Claude run.

6. The expert completed an evaluation form about usefulness, realism, trust, limitations, and requirements for practical adoption, see Appendix A.3

7. Additional comments from the session were noted where relevant.

3.1.3.4 Data Collection

The main data source for RQ3 was the evaluation form completed by the UI/UX expert after the evaluation session. The evaluation form was designed to keep the evaluation simple and focused on practical value.

The evaluation form covered the following areas:

- overall experience using the framework
- perceived usefulness for early-stage UI/UX validation
- quality and relevance of the generated insights
- perceived realism of the synthetic user behavior
- trust in the generated outputs
- obstacles or limitations encountered during use
- improvements required before the framework could be used in practice

In addition to the evaluation form, observations and comments from the evaluation session were used as supporting qualitative data. The AI-generated logs from OpenAI and Claude were treated as evaluation material shown to the expert, but they were not the primary object of analysis in RQ3. The primary object of analysis was the expert's perception of the framework after using and reviewing it. The full evaluation form and evaluation guide are provided in Appendix A.3

3.1.3.5 Evaluation Criteria

The evaluation was guided by five qualitative criteria.

Usefulness refers to whether the framework was perceived as helpful for early-stage UI/UX validation.

Insight quality refers to whether the generated outputs helped identify relevant usability issues or design considerations.

Behavioral realism refers to whether the synthetic user appeared to behave and reason in a way that resembled a real user.

Actionability refers to whether the generated outputs could support concrete design decisions or further investigation.

Practical applicability refers to what would still be required before the framework could be integrated into a real UI/UX workflow.

These criteria build on the earlier design goals of the framework and are consistent with the thesis focus on behavioral realism, insight quality, and practical usefulness.

3.1.3.6 Data Analysis

The evaluation form responses and supporting comments were analyzed using a qualitative descriptive approach. The analysis focused on identifying the expert's perceived strengths, weaknesses, and improvement needs for the framework.

Responses were grouped according to the evaluation criteria: usefulness, insight quality, behavioral realism, actionability, trust, and practical applicability. The analysis did not aim to produce statistically generalizable results. Instead, it aimed to provide an indicative assessment of whether the framework was considered useful by a practitioner and what improvements would be necessary before practical adoption.

The Claude run was analyzed only through the expert's comparative reflections. The purpose was to identify whether the expert perceived any meaningful differences between the OpenAI-based interactive run and the completed Claude run. No formal benchmarking between LLMs was conducted.

3.1.3.7 Deviation from Project Plan

The initial research design for RQ3 was based on a quantitative comparison between AI-based validation and human user testing, focusing on efficiency metrics such as time, cost, and scalability. However, during the development phase (RQ2), it became evident that the primary challenge was not automation efficiency, but rather the quality and realism of the generated outputs.

In particular, expert feedback highlighted that increased automation would not provide value if the generated insights were not sufficiently reliable or human-like. As a result, the evaluation focus was refined to prioritize qualitative assessment of behavioral realism, insight quality, and practical usefulness. This led to the adoption of a qualitative case study approach for RQ3.

3.2 Expected outcomes

This section presents the expected outcomes that guided the study before the artifact was implemented and evaluated. These outcomes define the intended technical and practical contributions of the research design.

3.2.1 Technical outcomes

The main technical outcome of this thesis is a functional AI-based validation framework designed to support early-stage UI/UX validation. The framework is capable of interacting with Figma-based designs or static images to perform predefined user tasks and generate real-time "think-aloud" logs based on persona-driven generative AI.

Rather than focusing on full automation, the framework prioritizes the quality, transparency, and interpretability of generated outputs. It produces structured artifacts, including stepwise interaction traces and "think-aloud" reasoning logs, which reflect how a simulated user navigates and evaluates a user interface.

The system follows a controlled interaction approach in which UI states are processed sequentially. This enables more realistic interaction behavior while ensuring that outputs remain verifiable and reproducible, supporting systematic evaluation. The framework also supports the simulation of multiple personas, enabling variation in behavior and reasoning across different user profiles.

The primary technical contribution lies not in the complexity of the system itself, but in the design of personas and prompt structures that guide the behavior of the

synthetic users. The study systematically explores how different persona definitions and prompting strategies influence behavioral realism, consistency, and the usefulness of generated outputs.

Throughout the development process, the framework is iteratively refined based on expert feedback to improve behavioral realism and alignment with professional UX evaluation practices. As such, the contribution extends beyond the implementation itself to include an exploration of how design choices in persona construction and prompt engineering impact the quality of AI-generated usability insights.

3.2.2 Value proposition

3.2.2.1 Practical Value

The expected practical value for Strawberry Planet was not assumed to lie primarily in the technical implementation of the framework, but rather in the guidelines and design principles for persona construction and prompt engineering that enable realistic simulation of GenAI-based users. While the framework facilitates the simulation of user behavior, the underlying technical implementation is relatively straightforward, consisting mainly of API calls and context management.

Instead, the study expected that the quality and usefulness of the generated insights would depend on how synthetic users are defined and guided. In particular, the design of personas and the structuring of prompts play a crucial role in achieving behavioral realism and producing meaningful, actionable feedback. This highlights that the primary contribution is not the complexity of the system itself, but the methodological knowledge required to effectively control and contextualize AI-generated behavior.

The framework was expected to contribute methodological knowledge rather than direct operational efficiency. It was expected that a more refined and fully developed version of the approach could provide practical value in addressing common challenges in early-stage UI/UX validation, such as time constraints, cost, and difficulties in participant recruitment.

Given that traditional validation processes are often resource-intensive and therefore conducted selectively, AI-based synthetic users could enable more frequent and lightweight validation during early design iterations. This suggests that, while the current contribution lies in the design principles for persona construction and prompt engineering, the approach has the potential to support more continuous and accessible validation practices in real-world settings.

3.2.2.2 Positioning of the Framework

The framework should be understood as a complementary decision-support tool rather than a replacement for traditional user testing. Its value is contingent on the perceived realism of simulated behavior and the usefulness of the generated insights within professional UX workflows.

The findings indicate that AI-based synthetic users may be particularly useful in early-stage validation scenarios where rapid iteration is prioritized, and where the cost and logistical constraints of human testing limit testing frequency. However, the

approach remains dependent on human judgment, especially in interpreting ambiguous or subjective usability issues.

3.2.2.3 Integration into Practice

The expected outcome was to explore how AI-generated feedback could be integrated in existing UX workflows as a complementary support tool. The Framework makes it possible to conduct early stage user validation tests for quicker and cost effective insights during early designs. At the same time this study establishes the boundaries were AI generated insights are sufficient and when human assessments are needed.

3.2.2.4 Research Contributions

The study contributes insights into how behavioral realism can be achieved in AI-based user simulations. It particularly highlights the importance of persona design and prompt structuring to guide AI behavior, as well as the trade-offs between automation, control, and quality of generated insights.

Chapter 4

Results and Analysis

4.1 Identification of Current Practices and Challenges (RQ1)

The objective of RQ1 was to investigate the practical landscape of early-stage UI/UX validation to identify current limitations and determine which activities offer the highest potential for Generative AI (GenAI) automation. This section presents the results of a thematic analysis derived from four semi-structured interviews using the Critical Incident Technique (CIT). The participants consisted of three UI/UX designers and one requirements analyst working closely with a team of UI/UX designers. The organizations consisted of two software consulting firms and one pure UI/UX consultant firm. The findings were organized into six primary themes: Workflow, Tools and Practices, Challenges / Pain points, What works well, Human-in-loop Boundaries, and Automation Candidates

First-Order Theme	Second-Order Categories (Sub-Themes)
Workflow	Design and Requirements engineering, Preparation, Internal sanity checks, External usability testing, Analysis and reporting, Validation strategies, AI usage in current workflow
Tools & Practices	AI tools, Recruitment, Designing, Validation testings
Challenges & Pain Points	Validation scope constraints, Recruitment constraints, Quality of insights, Logistical constraints, Internal sanity check limitations, time constraints, Cognitive load, AI limitations,
What Works Well	What works well when using AI, What works well when following manual workflows
Human-in-loop Boundaries	Interpretation of insights, Subjective UX judgment, Contextual understanding, Validation of AI outputs
Automation Candidates	Synthetic user / persona simulation, Script drafting, Result transcription, Prototype preparation and linking, Layout and visual hierarchy feedback, Behavioral flow analysis, Participant recruitment

Table 4.1: Overview of identified themes and sub-themes from thematic analysis

4.1.1 Workflow

The interviews revealed that early-stage UI/UX validation workflows vary between organizations depending on team size, project scope, and available resources. While the general structure of usability validation was similar across companies, specific practices and tools differed.

Because the artifact developed in this study is evaluated within the case organization Strawberry Planet, the workflow used there is presented as the primary reference workflow. Insights from the other interviewed designers are used to highlight similarities, variations, and common industry practices.

4.1.1.1 Design and Requirements engineering

Before any usability validation begins, designers first engage in design exploration and requirements analysis. At Strawberry Planet, this stage involves understanding the problem space, defining feature requirements, and developing initial design concepts. According to the interview with the company's UX/UI designer, this process typically includes research, sketching ideas, and creating early design drafts in Figma. During this phase the UX/UI designer at strawberry planet utilizes ChatGPT for quick peer review feedback, primarily on UI and not UX. Some internal sanity checks are done

with other employees at strawberry planet for validation on design and flow of the UX/UI. The designs in Figma often include UI elements for aesthetic purposes.

Across the other interviewed organizations, similar early design activities were described, although the structure of collaboration varied depending on team size, and the level of design varied during these early stages. Larger organizations have more UX/UI designers to work with and use as peer reviewers in comparison with strawberry planet removing the need of using GenAI tools such as ChatGPT for UI feedback. Internal sanity checks is also conducted with employees more closely related to the project. Requirements also pass through a lot more people and roles before reaching the designer. From a design standpoint one of the participants mentioned that a very early prototypes could be created using AI tools such as Lovable for early inspiration or problem identification. Another participant only created low level prototypes in black and white all the way until after the first round of user validation tests. This is to ensure that the test focus on UX elements more then UI. Some tools were also mentioned for exploring existing designs and inspiration from platforms such as, Dribble, Behance and Mobbin to help understand common design patterns and approaches used in similar products.

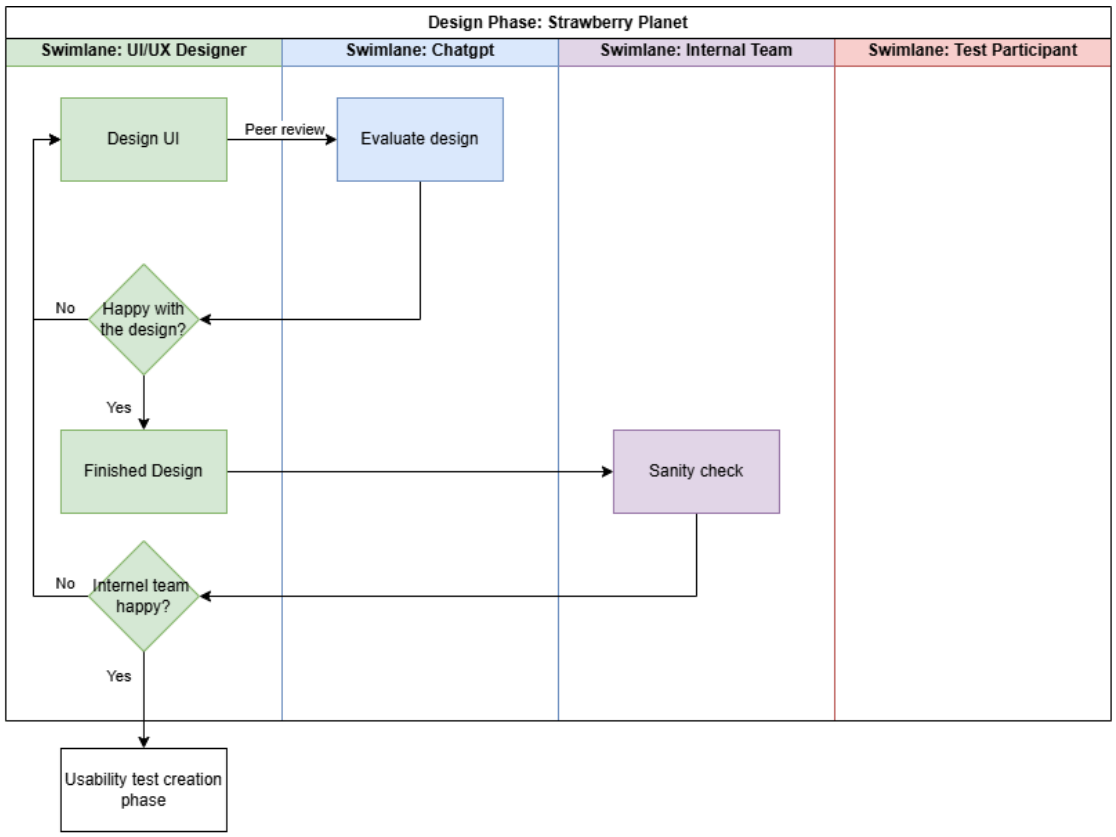


Figure 4.1: Design Phase

4.1.1.2 Preparation

Preparation in the context of Strawberry Planet consists of creating prototypes, identify features to validate, define validation goals, identify target audience, Recruitment

of participants, writing usability testing scripts and structuring tasks. Validation is typically only prioritized when a major new feature or workflow change is introduced rather than for small UI refinements. Preparation for a usability validation session can take several hours, including writing scripts, defining scenarios, structuring tasks, and linking prototype screens. Due to time constraints clickable prototypes are not always created and static images is used instead. Creating clickable prototypes requires manually connecting screens in Figma, which introduces additional setup work before testing can begin. Recruitment of participants are often very time consuming and hard to find the right audience.

The creation of validation goals, identifying target audience are often done by the UX designer in correlation with a requirement engineer or team leader. The creation of test scripts and structuring tasks was mostly done by the UX/UI designer that created the design. Recruitment of participants varied a lot between organizations. One organization used a tool called UserBrain that did all recruitment and conducted the usability tests for you, another consulting organization got its participant from the company that hired them. In some projects recruitment was done on the street with a prototype in hand, and other projects could follow the approach done at Strawberry Planet.

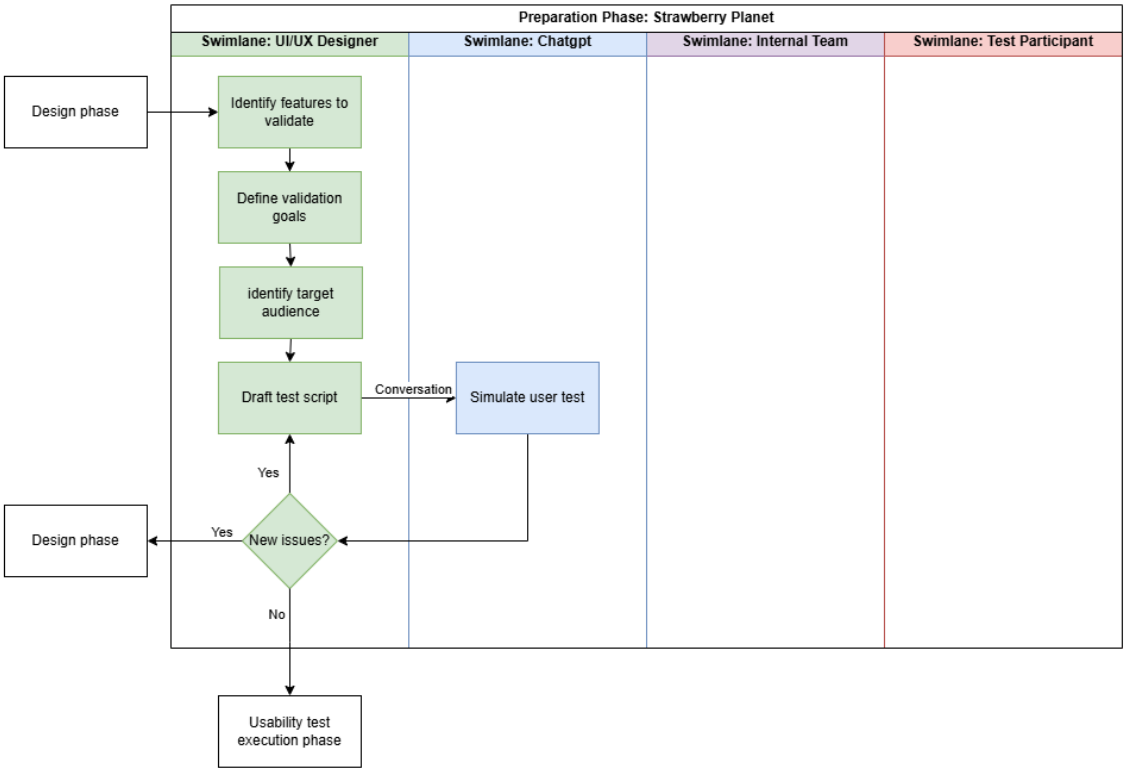


Figure 4.2: Preparation Phase

4.1.1.3 Internal sanity checks

Internal validation often involves quick reviews with developers or team members to verify feasibility and logic before external testing. Internal reviews primarily focus on feasibility or logic rather than full UX validation.

Internal feedback availability depends on team size and schedules, which can limit the depth of early feedback. The size of the organization also effect what roles are available for internal sanity checks. Larger companies with multiple UI/UX designer could do more internal checks regarding UX validation then smaller companies with a single designer.

4.1.1.4 External usability testing

External usability testing is the most formal part of the validation process. It consists of real human users interacting and testing the prototype to identify usability issues and to validate design decisions. At Strawberry Planet external usability testing is primarily conducted on larger new functionalities and not for smaller UI-changes. At Strawberry Planet the test are conducted remotely because of its geographical location and access of participants. A typical session starts with a introduction were the participant get some background and context of the application. After the introduction the participant execute some predefined tasks and utilizes the think aloud methodology. The tests often take around 40-45 minutes and end with a 5-10 minutes discussion with follow up questions about improvements and experiences. A common guideline is to conduct tests with at least five participants per iteration. The prototypes used are usually clickable Figma prototypes to enable realistic flows, although static screens are sometimes used instead due to time constraints. During the sessions, the UI/UX designer observes user behavior and documents insights through notes or recordings

In other organizations, external usability testing follows a similar structure, but with variations in implementation. Some organizations use specialized platforms that handle both recruitment and recording of tests, including automated transcription and highlighting features. Others conduct more informal testing, for example by recruiting participants spontaneously in public settings. There are also differences in the degree of moderation, with some tests being moderated and others being unmoderated with recording of user interactions. One organization strictly prohibits the designer from conducting the usability test, to minimize bias and give to many hints.

4.1.1.5 Analysis and reporting

The analysis and reporting phase focuses on transforming raw data from usability testing into structured and actionable insights. At Strawberry Planet, this process primarily consists of reviewing notes and any recordings to identify recurring patterns and usability issues. The analysis is often conducted shortly after testing is completed, as insights are easier to identify when sessions are close together in time. The process is largely manual and can be time-consuming, especially when multiple sessions need to be analyzed.

The results of the analysis are compiled in a report that typically contains:

- Identified usability issues
- Recurring patterns
- Prioritized improvements

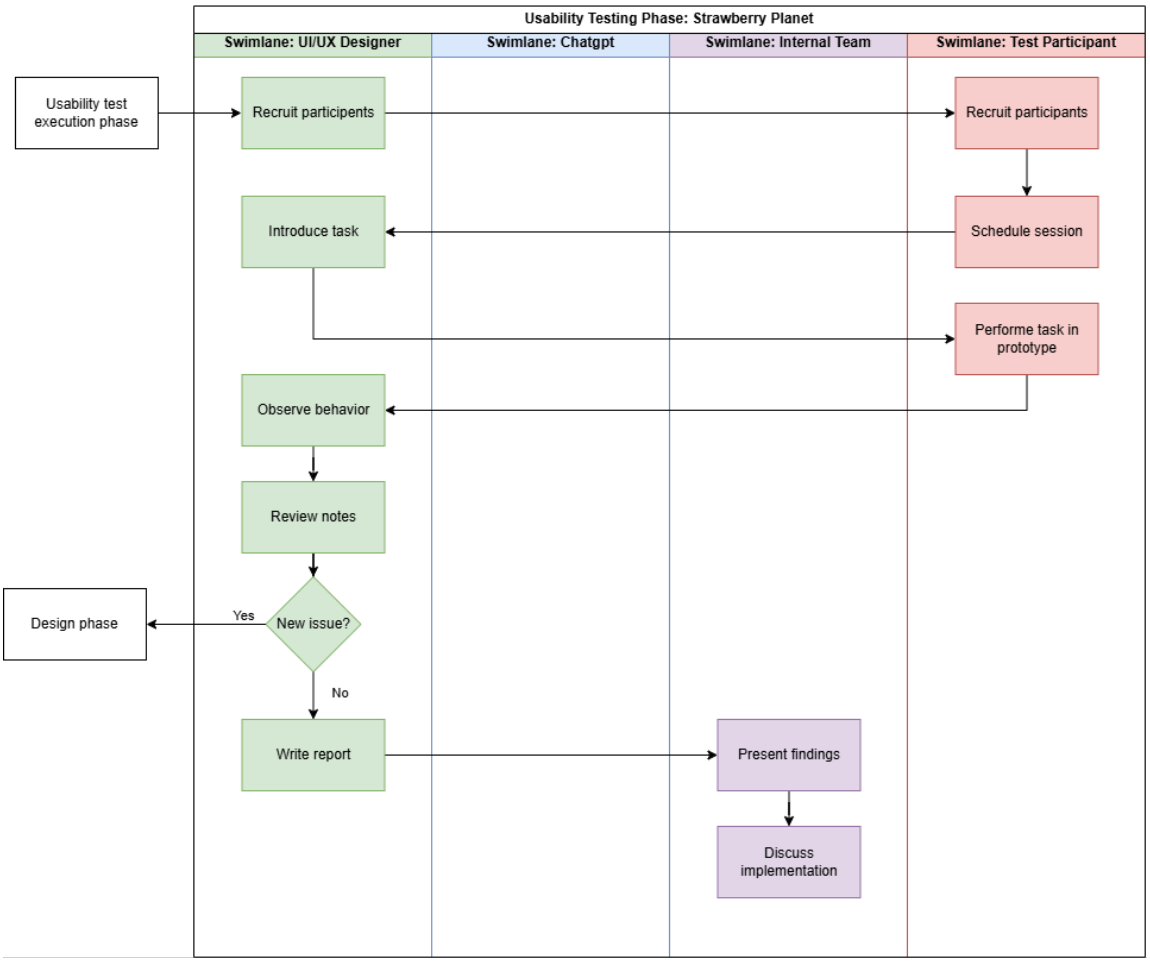


Figure 4.3: Usability Testing Phase

- Citations or observations by the user

Other organizations vary in the level of automation in this phase. Some use tools that offer automated transcription and AI-based summaries, which can streamline analysis and make it easier to navigate large amounts of data. However, manual interpretation is still required to ensure that insights are relevant and contextually accurate. In larger organizations multiple designer could both take notes during the test and analyze the findings afterwards.

4.1.1.6 Validation strategies

At Strawberry Planet they utilizes a clear *Shift-Left* methodology were the focus is on identifying issues as early as possible in the design process. Through the interview it was established that a combination of strategies were used, depending on what design and how far into the design process the prototype is. The primary strategy combine Flow-Based Validation and Task-Based Validation were the user gets a specific task in a *Clickable Figma prototype*. They are asked to think-aloud while there behavior and reasoning is observed. This makes it possible to analyze navigation logic, user expectation and decision points in the user flow. For smaller and earlier designs the focus is often more on Static Validation were layout, visual hierarchy and UI-elements are the most important. Strawberry Planet also utilizes some internal sanity checks were colleagues with different roles review the design before user validation testing is done.

Compared to other organizations, the overall strategies are similar, but with variations in how much detailed they are applied. Larger organizations tend to have more formalized validation strategies, where different levels of testing (e.g. low-fidelity vs. high-fidelity) are systematically used depending on the project phase. Some organizations consistently use low-fidelity prototypes (e.g. black and white wireframes) in early testing to isolate UX from UI, while Strawberry Planet more often includes visual elements early in the design.

4.1.1.7 AI Usage in Current Workflow

AI is already being used in Strawberry Planet's workflow today, but in a limited and supporting role rather than as a central part of the validation process. The most prominent use is that ChatGPT functions as a form of AI-based peer reviewer, where it is used to:

- Give quick feedback on UI-designs
- Propose improvements
- Helping with formulating test scripts

This compensates for the lack of more UX/UI designers on the team and enables rapid iteration in the design phase. At the same time, the importance of human-in-the-loop is emphasized, as AI output must be critically reviewed to avoid bias and incorrect conclusions.

Compared to other organizations, AI usage varies greatly. In larger teams, AI is used less for peer review because internal expertise is available. However, AI is used in some cases for:

- Generation of low-fidelity prototypes
- Inspiration Search
- Automating Simple Design Tasks
- Automated audio to text transcripts
- summarizing transcripts

In general, AI is not yet integrated into the User validation workflow itself to any great extent, but is mainly used as a support tool in the design and preparation phases.

4.1.2 Tools and Practices

4.1.2.1 AI tools

The thematic analysis revealed that Generative AI, specifically ChatGPT is already being integrated into the workflow as a "Synthetic Peer Reviewer." This practice has been seen particularly important for solo designers who lack immediate access to design colleagues for internal sanity checks. In this capacity, AI is used to provide early feedback on UI logic or to act as a support tool for identifying potential usability issues in static images.

4.1.2.2 Recruitment

Executing validation frequently relies a lot on remote usability testing platforms, which serve a dual purpose, recruiting participants from established pools and collecting recorded testing sessions. These platforms reduce the administrative burden by offering features such as automated transcription and timestamped feedback highlights, which allows designers to navigate long recordings more efficiently.

4.1.2.3 Designing

Figma is used as the primary design tool for UI design, prototyping, and interaction flow creation. Designers may also explore existing design inspiration sources before starting the initial mockups. They make use of single static designs, multiple static designs, and functional Figma prototypes (mostly during bigger and newer projects)

4.1.2.4 Validation testing

The validation testing methodology itself varies depending on the stakeholder involved. When conducting tests with real users, participants almost universally make use of the "think-aloud" method to capture real-time reasoning. However, early validation with internal stakeholders sometimes shifts focus away from layout and visual hierarchy. Instead, these early-stage reviews are used to verify domain correctness, legal requirements, or specific terminology.

4.1.3 Challenges / Pain points

4.1.3.1 Validation scope constraints

A significant finding within the validation workflow is the selective nature of testing, which can often be dictated by project scope and stakeholder pressure rather than a continuous quality assurance process. Participants noted that usability testing is frequently skipped because stakeholders prioritize visible deliverables, such as finalized UI designs over research activities. Because of this, external validation with real users are usually reserved for major new features or significant workflow changes, while the minor UI refinements are often just validated internally. Furthermore, the actual technical limitations of the early-stage material often hinder the depth in which validation can be done. Static prototypes are limited in what they can evaluate, lacking the ability to capture behavioral aspects, while complex, multi-screen flows remain difficult to test reliably without a fully functional prototype environment.

4.1.3.2 Recruitment constraints

The recruitment of suitable participants was identified as the most significant hurdle in the user validation process. Finding actual users who can accurately match the intended target group is a resource-heavy task, and even when utilizing professional participant pools, designers often encounter misalignment between the testers and the product context where testers do not fully grasp the product´s specific purpose or environment. The difficulty is even further increased when dealing with niche or sensitive user groups, for instance, certain populations are hard to reach due to accessibility barriers, lack of willingness to participate, or legal constraints such as GDPR. The quality of feedback is also often threatened by inherent participant biases. Long-term users of a product frequently bring an "expectation bias" to sessions, as their familiarity with the interface allows them to navigate around flaws that would impact a new user.

4.1.3.3 Quality of insights

The value of validation data is very different because of the variable quality of participant feedback. Designers reported that feedback relevance was heavily dependent on the individual participant, with some users providing shallow or surface-level comments that would fail to uncover any root usability issues. Another recurring analytical challenge is the difficulty in separating genuine function usability flaws from subjective personal preferences, which could lead to conflicting design directions. The lack of robustness could also be traced to having small sample sizes, which could lead to difficulties in generalizing the findings or reach a definitive consensus on a design.

4.1.3.4 Logistical constraints

The practical execution of user validation is very often limited by the need of remote testing. While it may help overcome geographic barriers, they also reduce the evaluator´s ability to observe subtle emotional or non-verbal reactions compared to

in-person sessions. Furthermore, remote testing can also introduce some technical friction during sessions, which can disrupt the flow of the validation test and distract from the primary goal.

4.1.3.5 Internal sanity check limitations

Internal validation is frequently used as a first step, but it often face several limitations. The effectiveness of these "sanity checks" depend heavily on the availability of the internal team, which can lead to delays in the design cycle. Additionally, feedback from developers often focus on technical feasibility rather than user experience. For these reasons, participants emphasized that internal validation is not a substitute for testing with real users.

4.1.3.6 Time constraints

Time constraints are a major factor in determining whether testing is conducted fully or not. The process in itself is very time-consuming, requiring significant effort for script creation, the execution of users tests, and the repetitiveness of session moderation. Furthermore, the analysis phase also remains very manual, as reviewing recordings to identify patterns is particularly labor-intensive, especially when multiple sessions are conducted. Participants also noted that current AI solutions can also be very time-consuming to manage.

4.1.3.7 Cognitive load

Moderating validation sessions can also introduce a high cognitive demand for the evaluator. This burden primarily stems from the challenge of actively maintaining participants engagement and moving the test forward. Balancing these tasks is perceived as mentally taxing, particularly when several sessions are conducted consecutively.

4.1.3.8 Ai limitations

The introduction of AI into validation introduces several technical and reliability challenges. The main concern being the risk of confirmation bias and over-reliance on AI suggestions, which requires a lot of human filtering of all outputs. This filtering process is necessary to critically assess the relevance, accuracy, and contextual validity of the generated insights before they can be used in decision-making. Current systems also lack emotional reactions and frequently struggle with complex, multi-step reasoning. Furthermore, the quality of the AI feedback is very reliant on prompt formulation, leading to a lack of reproducibility and significant session-to-session variability.

4.1.4 What works well

4.1.4.1 What works well when using AI

Despite the limitations previously noted, creating AI features are already perceived as effective when used in the current workflow. Specifically, AI-generated UI feedback

provides a valuable initial insight of critique on visual structure. AI has also been able to generate transcript summaries and summarize large sets of data which significantly reduces the time required for manual analysis. Using AI as a "peer reviewer" has also served as a practical solution for solo designers.

4.1.4.2 What works well when following manual workflows

The traditional aspects of the current validation workflow also provide several benefits. Clickable Figma prototypes are highly effective for simulating realistic navigation and interaction flows, allowing for testing before any development begins. Utilizing task-based testing is another successful practice, as it focuses on objective user behavior. Remote testing platforms are also valued for their ability to simplify recruitment and reduce the overall administrative workload for designers. Finally, the early collaboration between designers, developers, and requirements analysts is seen as a major benefit as it can help clarify misunderstandings before development.

4.1.5 Human-in-loop Boundaries

The thematic analysis identified multiple boundaries where human intervention remains essential. These boundaries define the practical limits of automation in early-stage UI/UX validation and highlight areas where AI-based support must remain supervised and interpretative rather than autonomous.

4.1.5.1 Interpretation of insights

On a technical level, there is a clear boundary regarding the interpretation of the insights. Designers still need to manually interpret usability findings to distinguish between actual usability issues and AI hallucinations.

This limitation arises because usability findings are inherently ambiguous and context-dependent. For example, a user hesitating before interacting with an element may indicate confusion, deliberate exploration, or uncertainty, and correctly interpreting such behavior requires domain knowledge and contextual understanding beyond what current AI systems can reliably provide.

4.1.5.2 Subjective UX judgment

The primary human responsibility identified was the nuanced interpretation of the behavior, while AI can simulate certain actions participants emphasized that the final decision-making power must always reside with the designer and that the AI-generated feedback is strictly viewed as a decision-support rather than a final conclusion or replacement.

UX evaluation often involves trade-offs between competing design goals, such as usability, aesthetics, and business requirements. These decisions require nuanced judgment and cannot be fully reduced to automated evaluation. As a result, AI-generated feedback is consistently viewed as decision-support rather than a replacement for human expertise.

4.1.5.3 Contextual understanding

Additionally, high-level business and strategic tasks such as requirements engineering are viewed as fundamentally human domains, requiring a level of contextual understanding that current automation cannot replicate.

Current AI systems lack the ability to fully integrate such broader context, limiting their effectiveness in tasks that depend on holistic reasoning and long-term design considerations.

4.1.5.4 Validation of AI outputs

Finally, participants highlighted the necessity of validating AI-generated outputs. This is particularly critical when working with static images or early-stage prototypes, where incomplete interaction context increases the risk of incorrect or misleading interpretations.

As a result, fully automated validation processes were viewed as unreliable without human oversight. Designers must therefore review and verify AI-generated insights before incorporating them into design decisions.

4.1.5.5 Summary

These findings indicate that while certain aspects of early-stage UI/UX validation can be supported by AI, core activities related to interpretation, judgment, and contextual reasoning remain dependent on human expertise. These boundaries directly inform the scope of feasible automation and support the adoption of a human-in-the-loop approach rather than fully autonomous validation.

4.1.6 Automation Candidates

The following automation candidates represent our interpretation of opportunities identified from the interview data, informal discussion with the participants after the interview and subsequent analysis of the interview results. They should therefore be understood as analytical conclusions rather than direct participant statements.

- **Synthetic user / persona simulation**: AI-based personas could execute predefined tasks on UI prototypes and simulate user reasoning through think-aloud logs. This could enable rapid early-stage UX validation without recruiting real participants.
- **Script drafting**: High automation potential. AI can generate initial usability testing scripts, task descriptions, and interview questions which can then be refined by designers.
- **Result transcription and summarization**: High automation potential. AI tools can automatically transcribe recordings and summarize insights, reducing manual analysis time.
- **Prototype preparation and linking**: Medium automation potential. Creating clickable prototype flows currently requires manual linking of screens, which could potentially be automated.

- **Layout and visual hierarchy feedback**: High automation potential. AI can reliably analyze static UI layouts and provide feedback on hierarchy, clarity, and visual structure.

- **Behavioral flow analysis**: Medium automation potential. AI may assist in identifying navigation issues and usability problems across multi-screen workflows, though complex flows remain challenging.

- **Participant recruitment**: Partial automation potential. Existing platforms provide tester pools, but synthetic users could further reduce reliance on recruiting human participants during early-stage validation.

4.1.7 Key findings

This section synthesizes the main findings from the thematic analysis, highlighting the most important patterns across the identified themes.

4.1.7.1 Validation is selective and resource-constrained

Usability testing is not conducted continuously, but rather selectively, mainly for major features or significant workflow changes. Smaller UI iterations often receive little or no external validation.

A primary reason for this is resource constraints, particularly related to time and participant recruitment. Finding suitable participants that match the target audience is difficult and time-consuming, which directly limits how frequently usability testing can be performed.

4.1.7.2 AI is used as a support tool rather than for validation

AI tools are already integrated into parts of the workflow, particularly in activities such as transcription of usability sessions, summarization of findings, and providing early feedback on UI ideas.

However, these tools are not used to perform validation itself. Instead, they primarily support analysis and improve efficiency within existing processes rather than replacing core validation activities.

4.1.7.3 Key challenges are operational rather than methodological

The current UX validation workflow relies heavily on manual processes, including participant recruitment, moderated test sessions, and manual analysis of insights. These activities require significant time and effort, especially for small teams or single UX designers.

The analysis phase is also time-consuming, as designers must review recordings, interpret think-aloud feedback, and manually identify patterns across sessions.

4.1.7.4 There is clear potential for partial automation

A key opportunity identified during the interviews is the potential use of AI-based synthetic users or personas to simulate user interactions during early-stage UI validation. Such synthetic users could execute predefined tasks on prototypes and generate reasoning logs similar to the think-aloud method used in traditional usability testing.

Synthetic users could reduce dependency on recruiting real participants and enable more frequent validation cycles during early design iterations.

4.1.7.5 Human judgment remains essential

However, several limitations were highlighted regarding AI-based testing, including the difficulty of simulating realistic human behavior, emotional reactions, and unpredictable interaction patterns.

As a result, synthetic users are most suitable as a complementary early-stage validation tool, supporting rapid exploration and iteration rather than replacing real user testing.

Human involvement remains essential for interpreting results, validating insights, and making final design decisions.

4.1.8 Conclusion to RQ1

The findings from RQ1 indicate that several parts of the early-stage UI/UX validation workflow could potentially be supported by GenAI, particularly activities related to persona simulation, task execution, and lightweight validation support. However, the identified automation opportunities were broad, and it was not yet clear which technical approach would be feasible or useful within the Strawberry Planet workflow.

Therefore, RQ2 begins by assessing the technical feasibility of using multimodal LLM's for screenshot-based UI interpretation. This is followed by an exploratory design phase where different levels of automation, controllability, setup effort, and designer involvement are compared before selecting a framework direction for iterative development.

4.2 Design and Refinement of the Synthetic User Framework (RQ2)

This section presents the design and refinement process of the AI-based synthetic user framework. The purpose of RQ2 was to investigate how such a framework could be developed to support early-stage UI/UX validation, with focus on behavioral realism, insight quality, and practical usefulness.

The section first presents the technical feasibility assessment, which examined whether multimodal LLMs could interpret UI screenshots sufficiently for the proposed approach. It then describes the design exploration phase, where alternative framework directions were compared, followed by the three prototype iterations and the cross-prototype analysis.

Together, these results explain how the final framework design emerged from both technical constraints and expert feedback, and why a controlled human-in-the-loop approach was selected instead of full automation.

4.2.1 Technical Feasibility

The technical feasibility of the proposed framework is dependent on whether current Large Language Models (LLMs) are capable of correctly identifying user interface elements from visual inputs. Since the framework relies on static UI screenshots as input, the ability to accurately identify and understand UI components is a prerequisite for simulating user interaction and reasoning.

To assess this, an initial feasibility test was conducted using three state-of-the-art multimodal LLMs: GPT, Claude, and Gemini. Each model was provided with UI screenshots and prompted to describe and interpret the visible interface elements, including buttons, input fields, navigation components, and visual hierarchy.

The feasibility assessment was conducted across multiple UI designs to reduce the risk of design-specific bias. One representative example is included in Appendix A.1. The observed patterns were consistent across the tested designs, particularly regarding the relative performance differences between the evaluated models.

The results show clear differences in performance between the models. Claude demonstrated the strongest capability in identifying UI elements and producing detailed and structured descriptions A.1.1. GPT performed similarly, although with slightly fewer elements and consistency A.1.3. In contrast, Gemini identified significantly fewer interface elements and produced less detailed outputs, making it less suitable for tasks requiring accurate UI interpretation A.1.2.

Despite Claude demonstrating the strongest performance, GPT was selected for the subsequent development phase. This decision was made to align the framework with the tools and workflows already used within the case organization. By using GPT, the development and evaluation were grounded in a realistic implementation context, ensuring that the results reflect practical applicability rather than an idealized setup.

Based on these findings, it can be concluded that the identification of UI elements is not a primary bottleneck for the proposed framework. While this does not establish the overall feasibility of the system, it confirms that current LLMs, particularly Claude and GPT, are capable of interpreting UI screenshots at a level sufficient to support further development and investigation.

4.2.2 Exploration Design

4.2.2.1 Goal of Exploratory Design Phase

During the early stages of development, it became evident that multiple architectural approaches could be used to implement the synthetic user framework. These approaches differed significantly in terms of automation level, controllability, and technical complexity. As a result, an exploratory design phase was conducted to identify and evaluate alternative solutions before committing to a final design direction. To guide this exploration, three key dimensions were defined as central to the

problem:

- **Automation level**: the extent to which user interactions with the UI prototype are automated.
- **Reliability and controllability**: the degree to which navigation flows and reasoning outputs can be verified and reproduced.
- **Setup effort**: the amount of preparation required from the designer before running a simulated usability test.

These dimensions reflect the core trade-offs between scalability, realism, and practical usability within a UI/UX workflow, and were used to structure the comparison of alternative designs.

4.2.2.2 Identified Design Alternatives

Based on the exploratory analysis, three design alternatives were identified and evaluated.

Option A – Full Automation. The first alternative represents a highly automated approach, where the system operates directly on a Figma prototype. The input consists of a prototype URL, a defined persona, and a task description. The system autonomously generates navigation flows, think-aloud logs, and a summarized report.

This approach offers a clean and scalable architecture, enabling controlled navigation transitions and high reproducibility across testing sessions. It also supports large-scale testing with multiple personas and simplifies the aggregation and evaluation of results.

However, the approach requires significant technical setup, including the preparation of fully interactive prototypes and additional implementation effort. This increases development complexity and makes it less feasible within the time constraints of the thesis. Furthermore, while the approach emphasizes automation, it introduces challenges in maintaining high-quality and realistic outputs.

Option B – Manual Loop Using Static Screens (Selected MVP). The second alternative adopts a controlled, human-in-the-loop approach based on static UI screens. The input consists of a design URL, a task description, and a persona, while the interaction flow is manually advanced by the designer through a stepwise process. The system generates think-aloud reasoning logs for each interaction step.

This approach provides high reliability and strong control over the interaction flow, as each step can be verified and guided by the designer. It also ensures reproducibility of reasoning outputs and minimizes technical risk, allowing the study to focus on evaluating the quality and usefulness of LLM-generated insights.

The primary limitation of this approach is the reliance on continuous human involvement, which increases designer effort and limits scalability when testing multiple personas. However, this trade-off allows for greater transparency and interpretability of the generated outputs.

Option C – Automatic Flow Inference From Static Screens. The third alternative explores an automated approach based on static screens without predefined navigation structures. In this design, the system attempts to infer navigation flows directly from UI screenshots.

This approach requires minimal setup effort and has the potential to function at very early stages of design, even before full prototypes are created. However, it introduces significant challenges. The inferred navigation flows are highly susceptible to hallucination, making it difficult to verify correctness. Additionally, the approach is technically complex and uncertain in feasibility, placing it outside the practical scope of the thesis.

4.2.2.3 Comparison of Design Alternatives

To systematically evaluate the three alternatives, a comparison was conducted across multiple dimensions, including automation level, reliability, setup effort, designer effort, scalability, output quality, technical risk, and applicability to early-stage design. The results of this comparison are summarized in Table 4.2

Dimension	**Option A**	**Option B**	**Option C**
Automation Level	High	Medium	High
Reliability	High	High	Low
Setup Effort	Medium	Low	Low
Designer Effort	Low	High	Low
Scalability	High	Low	High
Output Quality	Medium	High	Low–Uncertain
Technical Risk	Medium	Low	High
Works Early in Design	No	Yes	Yes

Table 4.2: Comparison of considered design alternatives

The comparison highlights clear trade-offs between the alternatives. **Option A** offers high automation and scalability but requires significant setup effort and introduces higher implementation complexity. **Option C** minimizes setup effort and supports early-stage use, but suffers from low reliability and high technical uncertainty due to the risk of incorrect flow inference.

In contrast, **Option B** prioritizes reliability, controllability, and output quality, while maintaining low technical risk. Although it requires higher designer effort and offers limited scalability, it provides a stable and transparent environment for evaluating the behavior and reasoning of synthetic users.

4.2.2.4 Decision / MVP

The decision to focus on a controlled synthetic user framework was partly informed by the RQ1 findings, where persona-based simulation was identified as a relevant automation candidate for early-stage UI/UX validation. However, the more specific decision to prioritize output quality, behavioral realism, and usefulness over a higher degree of automation emerged during the RQ2 design exploration. In other words, RQ1 motivated the general direction of automating parts of early-stage validation, whereas RQ2 clarified that automation alone would not be valuable unless the generated outputs were trustworthy and actionable for the designer.

Based on the comparison and identified trade-offs, Option B was selected as the basis for the Minimum Viable Product (MVP).

Although this approach limits scalability and requires continuous human involvement, it provides the highest level of reliability, transparency, and control over the interaction process. These properties are essential for evaluating the quality, realism, and usefulness of AI-generated insights in early-stage UI/UX validation.

This decision establishes the foundation for the subsequent prototype development described in Section 4.2.3 and directly influences the evaluation approach in RQ3.

4.2.3 Prototype 1

4.2.3.1 Design Goal

Prototype 1 was developed as an initial exploratory artifact to investigate the feasibility of simulating user interaction using a Large Language Model (LLM). The primary purpose of this prototype was to evaluate whether an LLM could successfully perform a reasoning–action loop within the context of UI/UX validation.

Specifically, the prototype aimed to assess whether an LLM is capable of:

- Interpret UI screenshots
- Follow a predefined task
- Simulate a persona
- Produce think-aloud reasoning
- Suggest structured interaction actions

The prototype served as an early proof-of-concept before investing in more advanced or automated solutions. Its purpose was to reduce uncertainty regarding whether synthetic user simulation is technically feasible and practically meaningful within the defined scope.

From a design perspective, Prototype 1 was implemented as a low-fidelity experimental system. It was developed as a TypeScript-based terminal application and deliberately focused on testing the core reasoning mechanism rather than full system automation. The interaction was semi-manual, maintaining a human-in-the-loop structure to allow controlled observation and evaluation of the LLM's behavior.

4.2.3.2 Implementation

Prototype 1 was implemented as a command-line interface (CLI) application using TypeScript and Node.js. The system operates through a structured interaction loop in which inputs are provided to the LLM and a constrained output is returned.

The core interaction loop consists of the following input and output structure:

- **Input:** UI screenshot, task description, persona definition, and limited interaction history
- **Output:** Structured JSON response generated by the LLM

The system relies on manual state transitions, meaning that navigation between interface states is handled by a human operator rather than automated logic. Logging functionality was kept minimal in this prototype. At the beginning of a session, the operator selects:

- A UI design (represented as a folder of screenshots),
- A predefined usability task,
- A persona describing the simulated user.

These inputs are combined into a prompt and sent to the LLM. For each interaction step, the LLM receives:

- The current UI screenshot,
- The task description,
- The persona description,
- A limited interaction history.

The full prompt structure used in Prototype 1, including task, persona, and instruction formatting, is provided in Appendix A.4.1. Based on this input, the model is instructed to:

- Interpret the interface,
- Decide on the next interaction step,
- Generate think-aloud reasoning explaining its decision.

The LLM response is constrained using a predefined JSON schema. The output contains three main components:

- Action (e.g., click, type, scroll, wait, done),
- Action parameters (e.g., targetText, textToType, scrollDirection),
- Reasoning, which represents the think-aloud explanation of the model's interpretation and decision-making process.

The system maintains a limited interaction context, only the two most recent interaction steps are included, earlier reasoning is not retained in the active prompt. This reduces prompt size and complexity while maintaining minimal continuity between steps. A separate logging mechanism records, the selected action, action parameters, generated reasoning and all interaction steps are stored sequentially. After task completion, the system compiles these logs into a usability transcript representing the full interaction session.

4.2.3.3 Observed Behavior

During the execution of Prototype 1, several behavioral patterns and system characteristics were observed that informed the subsequent design iterations.

The LLM was able to successfully perform the core interaction loop. It could interpret UI screenshots, follow predefined tasks, generate think-aloud reasoning, and produce structured action outputs. This confirmed that the fundamental reasoning–action loop was technically feasible within the defined setup.

The generated outputs were consistent in structure due to the enforced JSON schema, which enabled clear separation between actions, parameters, and reasoning. The think-aloud reasoning provided traceability for the model's decisions and allowed step-by-step inspection of the interaction process.

Persona input influenced the generated reasoning to some extent. However, the variation in behavior between different personas was limited.

The interaction process relied on a stepwise loop where each decision was made based on a single UI state and limited interaction history. This resulted in a sequential navigation pattern, where each step was dependent on the immediate input rather than a broader understanding of the full interface structure.

The system required continuous human involvement for selecting subsequent UI states. This created a tightly coupled interaction between the operator and the AI, where the operator controlled navigation while the AI generated reasoning and suggested actions.

Additionally, the outputs were sensitive to prompt formulation, meaning that small changes in task descriptions or persona definitions could influence the generated behavior.

The observations from Prototype 1 confirmed that the core system components were functioning as intended, while also revealing limitations related to persona influence, interaction flow, and system control.

4.2.3.4 Issues

The observations from Prototype 1 led to the identification of several design-related issues that informed the development of subsequent prototypes.

A primary issue was the reliance on manual state transitions. The system depended on a human operator to interpret the LLM's suggested action and select the next UI screenshot. This limited the level of automation and introduced additional effort in the interaction process.

Closely related to this was the absence of system-managed navigation logic. Since the navigation between UI states was not handled by the system, the LLM was unable

to independently manage or infer interaction flows across multiple screens.

Another issue concerned context management. The system only retained a limited interaction history, which constrained the amount of information available to the LLM for each decision step. This affected how previous actions and reasoning were incorporated into subsequent steps.

The prototype also revealed limited control over persona behavior. While personas were included as part of the input, their influence on the generated outputs was not strong or consistent, indicating a need for more structured persona definitions or improved prompt design.

Finally, the overall interaction setup required significant human responsibility, including defining tasks, creating personas, preparing UI flows, and interpreting outputs. This distribution of responsibilities limited the level of autonomy of the system and indicated areas where further system support could be explored.

These issues highlight key areas for improvement in terms of automation, control, and interaction design, and directly informed the changes introduced in subsequent prototypes.

4.2.3.5 Changes

The findings from Prototype 1 reinforced an important design decision that had already emerged during the exploratory design phase: increasing automation would not provide value unless the generated outputs were sufficiently useful. While Prototype 1 confirmed the technical feasibility of the reasoning–action loop, it also made clear that feasibility alone was not enough. The more important next step was therefore not to maximize automation, but to improve the quality of the generated outputs.

Following discussion with the UI/UX expert, the development focus shifted toward improving the usefulness of the simulated interaction rather than further automating the interaction flow. This included a need for stronger persona control, better behavioral guidance, and greater variation in the generated outputs. In addition, the next iteration was expected to support interaction during execution, such as allowing questions during the session and follow-up questions after the test.

4.2.4 Prototype 2

4.2.4.1 Design Goal

The primary design goal of Prototype 2 was to extend the synthetic user framework beyond static, stepwise execution by introducing interactive capabilities during task performance. While Prototype 1 focused on generating think-aloud reasoning for predefined UI states, it lacked mechanisms for dynamic exploration or clarification during the simulation.

Prototype 2 was therefore designed to enable a more flexible and exploratory interaction process, where the UI/UX designer could engage with the synthetic user during execution. This was intended to better reflect real-world usability testing practices, where participants are often asked follow-up questions and encouraged to elaborate on their reasoning.

A secondary goal was to improve the structure and consistency of outputs, making them easier to interpret and evaluate. This included refining both the prompt

structure and the format of generated responses, with the aim of supporting clearer reasoning traces and more actionable insights.

4.2.4.2 Implementation

Prototype 2 introduces several structural and behavioral changes to the framework, focusing on improving control over the synthetic user's reasoning, increasing persona fidelity, and enabling interactive runtime intervention.

Context Handeling and Runtime Inputs. Compared to earlier iterations, Prototype 2 expands how contextual information is handled during execution. In addition to processing individual UI states, the system now maintains and forwards:

- previous reasoning steps
- prior actions
- intervention inputs from the designer

This information is included in subsequent prompts, allowing the model to maintain continuity across interaction steps. The system therefore moves from isolated step execution toward a context-aware interaction process, where earlier decisions influence later reasoning.

At runtime, two explicit interaction modes are introduced:

- **Chat Steer:** Allows the designer to influence the model's next decision. This is used, for example, when the prototype does not fully support certain interactions (e.g., text input), or when the model fixates on incorrect assumptions.
- **Chat Explain:** Allows the designer to ask follow-up questions about the model's reasoning, enabling deeper inspection of decision-making and clarification of unexpected behavior. More closely representing real user validation sessions.

Prompt Structure (INTENT / DOMAIN / DEMAND) The prompt structure used in Prototype 2 (see Appendix A.4.2) is reorganized into a more rigid hierarchy consisting of three explicitly labeled components:

- **INTENT:** Defines the overall task goal and what the synthetic user is expected to achieve.
- **DOMAIN:** Describes the persona's knowledge base, background, and behavioral context.
- **DEMAND:** Specifies constraints on how the model should behave, including reasoning style, tone, and output format.

This structuring is consistent with prompt engineering practices where separating task intent, contextual grounding, and constraints helps reduce ambiguity and improve output consistency. [12]

Furthermore, clearly defining domain and demand has been shown to reduce the tendency of LLMs to produce generic or overly assistant-like responses by anchoring outputs to a specific context and set of constraints. [16]

Persona Design and Behavioral Guidance Persona design is significantly expanded in Prototype 2. Instead of brief role descriptions, personas are defined using more detailed attributes, including:

- demographic characteristics (e.g., age, occupation)
- qualitative traits (e.g., anxious, impatient)
- behavioral tendencies and preferences

This represents a structural shift from simple role descriptions toward behavioral guidance mechanisms, where the persona actively shapes how the model interprets and interacts with the UI.

In addition, personas are supported with demonstration examples, including examples of reasoning and hesitation. This corresponds to few-shot prompting approaches, where providing examples in context can help guide model behavior and improve consistency. For example "If a screen lacks a clear primary button or has too many competing elements, you feel a wave of mental fatigue." [12]

Reasoning Control and Think-Aloud Enforcement Prototype 2 introduces explicit control over how reasoning is generated during execution.

Both referenced sources highlight the importance of Chain-of-Thought (CoT) prompting, where step-by-step reasoning improves the model's ability to perform complex tasks. [12] [16]

To operationalize this, the prompt enforces a strict Think-Aloud protocol using a structured reasoning trace. The model is explicitly instructed to:

- "think step-by-step"
- "State what you notice before acting"

This requires the model to:

- describe observed UI elements
- explain intermediate reasoning steps
- justify actions before execution

The intention is to approximate how human users verbalize their reasoning during usability testing.

Post-Session Processing. After task execution, the prompt includes instructions for generating a final evaluation based only on the reasoning produced during the think-aloud steps.

This introduces a form of internal feedback, where the model reflects on its own prior outputs when producing a final summary. Such approaches have been identified as a way to improve the consistency and accuracy of generated outputs. [16]

Additionally, a tone requirement is introduced for the final output, instructing the model to align its language with the assigned persona (e.g., "sound like a retiree, not an AI analyzer"). This constraint is intended to prevent the model from reverting to a generic assistant tone and to maintain consistency with the defined persona.

4.2.4.3 Observed Behavior

The introduction of structured prompting, enhanced persona design, and interactive runtime capabilities in Prototype 2 resulted in several observable changes in how the synthetic users behaved during task execution.

A primary observation was the effect of the enforced think-aloud reasoning structure. Compared to earlier iterations, the synthetic users consistently produced more explicit descriptions of what they observed in the interface before taking action.

Another observed change was related to the expanded persona design. The inclusion of more detailed persona attributes, such as background, occupation, and qualitative traits, resulted in variations in reasoning style and decision-making across different runs.

At the same time, the influence of the prompt structure and instructions remained visible in the generated behavior. In several cases, the reasoning appeared to follow a structured and systematic pattern that was consistent across personas, indicating that the output was partly shaped by the imposed reasoning format rather than solely by persona-specific characteristics.

The use of demonstration examples (few-shot prompting) influenced how reasoning was expressed, but also introduced unintended effects. When examples were closely tied to specific UI details, the synthetic users sometimes produced responses that appeared overly guided by these examples. In some cases, this led to the identification of issues that were not clearly grounded in the current UI state, suggesting that the model was influenced by patterns present in the examples rather than solely by the observed interface. This resulted in reasoning that could appear forced or biased toward the provided examples.

The introduction of interactive runtime input influenced how reasoning evolved during execution. Through Chat Explain, the synthetic users were able to expand on their reasoning and clarify decision-making when prompted. Through Chat Steer, the designer could influence the direction of the interaction, for example by correcting misinterpretations or guiding the model when the prototype lacked full functionality. This interactive capability enabled the exploration of additional insights beyond the predefined task flow, allowing specific UI elements and decisions to be examined in greater detail through follow-up interaction.

The observed behavior indicates that Prototype 2 produced more structured and transparent reasoning processes, while also showing that the generated behavior was influenced by both persona definitions and the underlying prompt structure. This

resulted in outputs that were consistent and explainable, but not always fully aligned with how real users would interpret or react to the interface.

4.2.4.4 Issues

Several issues were identified in Prototype 2 related to behavioral realism, prompt influence, and interaction dynamics. To verify that these issues came from the LLM's behavior rather than artifacts of the technical implementation, the raw API prompt payloads were regularly reviewed to confirm that all of the instructions and constraints were correctly sent to the model.

A primary issue was the influence of prompt structure on behavior. Despite the expanded persona design, the generated reasoning often followed a consistent and structured pattern across different personas. This indicates that behavior was partly driven by the imposed reasoning format and instructions, rather than fully emerging from persona-specific characteristics.

The use of few-shot prompting introduced additional challenges. When examples were closely tied to specific UI scenarios, the synthetic users sometimes produced responses that appeared guided by these examples rather than the current interface. This occasionally resulted in the identification of issues that were not clearly grounded in the presented UI, reducing the reliability of the generated insights.

Another issue was the system's limited ability to capture user expectations and misunderstandings. The synthetic users did not consistently express mismatches between expected and actual system behavior, even when such discrepancies had been observed in real user testing.

Inconsistencies were observed in how strictly personas were followed. In some cases, explicit persona constraints were not respected, such as language limitations (e.g., personas defined as only understanding Swedish still producing responses beyond that scope). This indicates that persona definitions were not consistently enforced during execution.

These observations raised a central concern, particularly among stakeholders at Strawberry Planet: to what extent the AI was actually adhering to the defined personas. While the prompts included persona descriptions and behavioral guidance, it remained difficult to determine whether the generated reasoning and actions were genuinely influenced by these definitions, or primarily driven by the model's default behavior. This uncertainty made it challenging to assess the reliability of persona-based simulation within the framework.

4.2.4.5 Changes

Based on the identified issues in Prototype 2, several changes were defined to improve persona control, behavioral realism, and trust in the generated outputs.

A primary focus was to improve confidence in persona-driven behavior. To address the uncertainty of whether outputs were influenced by the persona or the underlying model behavior, the design direction for the next iteration emphasized clearer separation between persona characteristics and system instructions.

To make differences between personas more observable, the personas were planned to be made more distinct and extreme in their attributes. The intention was to create

clearer variation in behavior across simulation runs, making it easier to assess whether outputs were consistently driven by the defined persona rather than general model tendencies or task structure.

Finally, the role of prompt structure and examples was identified as an area for refinement. Adjustments were planned to reduce unintended bias from few-shot examples.

These changes drove the design goals of Prototype 3, where the focus shifted toward strengthening persona influence and increasing trust in the generated behavior.

4.2.5 Prototype 3

4.2.5.1 Design Goal

The primary design goal of Prototype 3 was to address the behavioral inconsistencies and trust issues identified in Prototype 2. While Prototype 2 successfully introduced interactive elements and structured reasoning, the generated personas still occasionally exhibited overly analytical behaviors, acting more like UX evaluators than real users.

Prototype 3 was therefore designed to completely separate the system's operational rules from the simulated user's mindset. The goal was to try and ensure that the persona's choices and reasoning were driven by real human psychological traits, increasing trust in the generated outputs and ensuring distinct divergence between the different synthetic users.

4.2.5.2 Implementation

Prototype 3 introduces a fundamental restructuring of the persona engine, shifting from descriptive text prompts to parameterized psychometric models. The full prompt structure, including persona definitions and behavioral constraints, is provided in Appendix A.4.3

Psychometric Framework (BFI-2). The persona definition was formalized using the BFI-2 Expanded psychometric framework (Big Five / OCEAN). Instead of relying on qualitative adjectives, each persona was explicitly defined through trait values across five dimensions:

- Open-Mindedness
- Conscientiousness
- Extraversion
- Agreeableness
- Negative Emotionality

This change transformed personas from simple role descriptions into systematic, repeatable behavioral models. The prompt structure was updated so that these traits directly influenced the LLM's decision-making style, its reactions to UI elements, and its tolerance for ambiguity.

To operationalize the psychometric framework, each trait dimension was translated into explicit behavioral tendencies within the prompt. For example, high negative emotionality was associated with hesitation and fear of making irreversible errors, low open-mindedness with distrust toward unfamiliar or abstract interface elements, and high conscientiousness with slower, more careful interaction patterns. This ensured that personality traits did not remain abstract descriptors, but actively shaped the reasoning and decision-making process of the synthetic user.

Real-World Few-Shot Prompting. In Prototype 2, few-shot examples were tied to UI/UX scenarios, which caused the LLM to adopt UX terminology often referring to the examples even if it wasn't relevant. In Prototype 3, this was changed. Few-shot examples were rewritten to depict everyday, non-digital scenarios directly connected to their personality traits (e.g., how a highly "Negative Emotionality" persons reacts to a confusing microwave interface). This forces the LLM to internalize the psychological trait rather than trying to mimic a design critique.

In addition to positive behavioral guidance, the prompt design introduced explicit negative constraints to suppress default AI assistant behavior. The model was instructed not to use UX/UI terminology, not to rationalize unclear interface elements, and not to adopt an expert perspective. Instead, it was required to remain within the persona's knowledge and express confusion, hesitation, or uncertainty when appropriate. These constraints were essential to maintaining realistic, non-expert behavior and preventing the model from defaulting to analytical or solution-oriented responses.

Pre-Context Injection. A new input option was introduced to provide the LLM with prerequisite domain knowledge before the test began. If a real user would logically understand domain-specific terminology within the application, this information was injected into the prompt to prevent the synthetic user from experiencing unrealistic semantic confusion.

This became particularly important in cases where terminology carried specific meaning within the product context. For example, in the Strawberry Planet application, the term "meeting" does not refer to a generic calendar event, but to a structured, guided collaborative session embedded within a "boost"-a multi-step development journey. In earlier iterations, the synthetic users interpreted "meeting" using its general meaning, which resulted in responses that did not align with real user behavior.

To address this, a dedicated product context section was introduced in the prompt, explicitly defining key concepts such as "boost" and "meeting" and explaining their role within the application workflow. This ensured that the synthetic user operated with the same conceptual understanding as a real user.

Providing this contextual grounding significantly improved the realism of the generated responses. The synthetic users were better able to interpret interface elements correctly and produced outputs that more closely resembled those of real users interacting with the system.

Extreme Personas. To further investigate the influence of persona design on generated behavior, the development process included the creation of deliberately "ex-

treme" personas. These personas were defined using exaggerated trait values within the selected psychometric framework, resulting in more pronounced and distinctive personality profiles.

The purpose of introducing such extreme personas was not to represent realistic user distributions, but to amplify behavioral differences and make the effects of persona-driven prompting more observable in the generated outputs. By increasing the contrast between personas, it became easier to identify whether variations in reasoning, decision-making, and interaction patterns were actually driven by the persona definitions rather than by inherent model bias or randomness.

This approach enabled clearer differentiation between outputs across simulation runs and provided a form of stress-testing for the framework. In particular, the extreme personas exposed edge cases in behavior, such as overly cautious, highly impulsive, or inconsistent reasoning patterns, which would be less visible in more moderate persona configurations.

While these exaggerated profiles may not fully reflect realistic end users, they played an important role in validating the underlying assumption of the framework: that structured persona definitions can meaningfully influence AI-generated behavior. The findings from these tests informed subsequent refinements toward more balanced and realistic persona configurations.

4.2.5.3 Observed Behavior

During the execution of Prototype 3, the following behavioral patterns were observed:

Persona Divergence: The BFI-2 parametrization successfully produced distinct viewpoints. When processing the exact same UI task (a Team Reflection Survey), different personas generated completely different usability friction. For example:

- A highly anxious persona (Annica) read all text and focused on the fear of data loss, requesting confirmation dialog.
- An impatient persona (Maya) ignored text, clicked rapidly, and requested a "Skip All" button.
- A highly conscientious persona (Victor) audited the structure and criticized the use of emojis, requesting a formal Likert scale.

Reduction of AI-Tone: The use of real-world few-shot prompting effectively stopped the LLM from using UX jargon. The reasoning traces felt significantly more naturalistic and connected to the assigned persona.

Edge-Cases: To stress-test the system's behavioral constraints, the prototype was run using highly polarized personas. The system successfully maintained extreme personality constraints (e.g., extreme anxiety vs extreme extraversion) without reverting to a standard helpful assistant mode.

Context Utilization: The pre-context injection appeared to reduce irrelevant confusion caused by product-specific terminology. By defining important platform concepts before the task began, the synthetic users were less likely to misunderstand the basic premise of the design. As a result, more of the think-aloud reasoning focused on the actual UI navigation, interaction flow, and layout, rather than on confusion about domain-specific words.

This was particularly useful because some terms in the tested design had meanings that were specific to the product context. Without this context, earlier outputs sometimes interpreted these terms too generally, which created confusion that would not necessarily occur for a real user familiar with the product. The added context therefore helped align the synthetic user's starting knowledge with the intended test scenario.

Well-Known Fictional Character Personas. As a final exploratory step in this phase, an additional experiment was conducted using well-known fictional character-based personas. The initial idea was to test characters from Winnie-the-Pooh because they represent broad and clearly distinguishable personality types, such as anxious, energetic, pessimistic, or calm behavior. This made them useful as stress-test personas for observing whether the framework could maintain strong behavioral differences across the same UI task. The results indicated more clearly differentiated behavior compared to earlier persona configurations, suggesting that recognizable persona archetypes may help strengthen behavioral consistency. This was not based on a formal metric, systematic coding process, or independent expert judgment. The observation should therefore be interpreted as an exploratory indication that recognizable persona archetypes may strengthen behavioral variation, rather than as a validated evaluation result.

This experiment was exploratory and was not treated as a formal evaluation result. The observation is therefore included here as an indication that persona recognizability may influence the strength and consistency of generated behavior, while the broader implications are discussed further in Chapter 5

4.2.5.4 Issues

Several issues were identified in Prototype 3 related to trust, behavioral realism, and the difficulty of verifying whether the model fully followed the assigned persona constraints.

A primary issue was that the AI remained a black box. Although Prototype 3 made the persona instructions more explicit and produced more traceable think-aloud outputs, it was still not possible to verify exactly how the model interpreted or applied the persona internally. The generated reasoning could show whether the output appeared consistent with the persona, but it could not prove that the model's decisions were actually caused by the persona constraints.

This issue became visible in relation to language understanding. In one test, a persona was instructed to only understand Swedish while the UI was presented in English. Even if the generated output could reflect confusion or limitation on the surface, it remained unclear whether the model actually simulated limited English understanding during the decision process, or whether it internally interpreted the

English UI and then adjusted the response afterward. This created uncertainty about how reliably such constraints were followed.

Persona adherence was improved compared to earlier prototypes, but it was not fully guaranteed. The stronger persona definitions helped create more visible differences between users, but the model could still drift toward coherent, helpful, or overly logical reasoning.

A related issue was that stronger personas could also exaggerate behavior. Making personas more distinct helped reveal differences between simulated users, but it also introduced the risk that the behavior became too extreme or stereotyped.

Taken together, these issues show that Prototype 3 improved behavioral differentiation and reduced some earlier problems related to generic AI-like responses. However, important limitations remained regarding trust, realism, and verifiability.

4.2.6 Cross-Prototype Analysis

4.2.6.1 Evolution Table

The evolution across prototypes primarily reflects improvements in control, prompting, and persona design, rather than changes in system architecture. This highlights that the effectiveness of the framework is driven more by behavioral guidance than by technical implementation complexity.

Table 4.3: Evolution of the prototypes across key design dimensions

Dimension	**Prototype 1**	**Prototype 2**	**Prototype 3**
Primary goal	Validate the feasibility of screenshot-based synthetic user interaction	Improve reasoning quality, interaction control, and persona guidance	Improve persona consistency, behavioral differentiation, and practical usefulness
Core architecture	CLI-based tool using screenshot input, task input, persona input, and JSON output	Same core architecture retained, with added interactive runtime functions	Same core architecture retained, with refined prompt and persona structure
Interaction model	Human-in-the-loop, with manual selection of the next UI state	Same stepwise model, supported by Chat Explain and Chat Steer	Same stepwise model, with clearer separation between system rules and persona behavior

Prompt structure	Basic prompt combining task, persona, image, and limited interaction history	More structured prompt with enforced think-aloud reasoning, constraints, and few-shot examples	Further refined prompt hierarchy with system rules, persona mindset, negative constraints, and pre-context
Persona design	Simple persona descriptions with limited behavioral influence	Expanded personas with demographic attributes, qualitative traits, and behavioral tendencies	BFI-2-inspired persona parametrization with stronger behavioral constraints and clearer persona divergence
Model output	Structured JSON output containing action, action parameters, and reasoning	Same JSON structure, with more explicit think-aloud reasoning and improved transparency	Same output structure, with more persona-specific reasoning and clearer variation across personas
Control over AI behavior	Low – the model often remained generic, overly rational, or assistant-like	Medium – control improved through structured prompting, but prompt influence remained visible	Higher – persona behavior became more distinguishable, but full adherence could not be verified
Logging and traceability	Basic sequential logging of actions, parameters, reasoning, and interaction steps	Improved traceability through more explicit reasoning and interactive clarification	More traceable persona-driven outputs, but internal model reasoning remained a black box
Observed strengths	Confirmed that the reasoning-action loop was technically feasible	Produced more structured, transparent, and explainable reasoning	Produced stronger persona divergence, reduced AI-tone, and more task-focused reasoning
Observed limitations	Limited persona variation, prompt sensitivity, limited context, and continuous human involvement	Reasoning was still shaped by prompt structure; few-shot examples sometimes biased the outputs	Remaining limitations in realism, trust, verifiability, and possible exaggeration of strong personas

Key design insight	Feasibility was confirmed, but output quality and persona influence were insufficient	Output quality depended heavily on prompt structure, examples, and persona design	Stronger persona structure improved differentiation, but human interpretation remained necessary

4.2.6.2 Key Insights

The cross-prototype comparison shows that the main improvements were not caused by major technical changes, but by increased control over the AI's reasoning, persona behavior, and output structure. The basic interaction loop remained similar across the prototypes, but the generated results became more useful as the prompts and persona definitions became more specific.

A key insight from Prototype 1 was that technical feasibility alone was not enough. Although the model could interpret UI screenshots and suggest actions, the outputs were often too generic or overly rational. This showed that the main challenge was not only to make the AI understand the interface, but to make it behave like a limited and subjective user.

Prototype 2 showed that structured think-aloud reasoning improved transparency, because the designer could inspect why the AI suggested a certain action. However, clear reasoning did not automatically mean realistic reasoning. The model could still behave more like an evaluator than a real user.

Prototype 3 showed that stronger persona definitions improved behavioral variation. The use of more specific constraints and BFI-2-inspired persona traits made the synthetic users more distinguishable. However, persona adherence remained difficult to verify, since the model's internal reasoning could not be directly observed.

4.2.7 Final Prompt Design

The full final prompt structure is provided in Appendix A.4.3. An example of a completed Prototype 3 user test run is provided in Appendix A.5.

1. Base System Constraints: The input begins by establishing the operational boundaries for the LLM. It forces the AI to act as a UI evaluator responding exclusively with a strict JSON schema. The AI's capability to make decisions on the UI is bound to a specific set of actions (click, type, scroll, wait, and done), and it is mandated to output a stream-of-consciousness reasoning trace to explain its decisions.

2. The Chosen Task: Next, the input injects the specific validation goal. To ensure the synthetic user has the same baseline understanding as a real user, this module also has the option to utilize Pre-Context Injection to define domain-specific terminology (e.g., explicitly defining what a "boost" or a "meeting" means within the company context). It also establishes two execution states:

- State 1 (Ongoing Navigation): The LLM must filter its observations entirely through the persona's traits before selecting its next option.
- State 2 (Final Reflection): Triggered only when the action 'done' is selected. The LLM must output a retrospective summary addressing the bullet points decided by the designer for that specific task.

3. The Chosen Persona: Finally, the system appends the parameterized psychological profile, Using the BFI-2 framework, the persona (e.g., "Annica") is injected:

- Profile & Traits: Variables like High Conscientiousness and Low Agreeableness force the model to prioritize efficiency and express irritation at decorative "fluff" or ambiguous UI elements.
- Cognitive Examples: Non-digital, real-world few-shot examples anchor the psychological traits without relying on UX jargon.
- The Constitution: Strict negative constraints forbid professional UI terminology (e.g., "visual hierarchy") to prevent the AI from defaulting to an expert.

(The full technical implementation of this final prompt architecture, including all system constraints and persona configurations, can be reviewed in A.4.3

Mid-Test State Change: Interactive Chat Mode To replicate the dynamic nature of human-led usability testing, the architecture allows the test to be interrupted. If the researcher pauses the test to ask a question, the system inject a state change (STATE CHANGE: MODERATOR INTERVIEW) into the ongoing context window. This overrides the JSON contract, forcing the LLM into a direct, first-person text conversation. Based on the researcher's input, the prompt enforces two conditional modes:

- Explain Mode: The synthetic user answers questions about its past actions and internal reasoning.
- Steer Mode: The synthetic user accepts new scenarios or guidance without advancing the UI state.

By keeping this within the same test session, the synthetic user maintains complete memory of the prototype flow while remaining strictly in character.

4.2.8 Summary of RQ2 Results

RQ2 asked how an AI-based synthetic user framework utilizing personas can be designed and refined to effectively support early-stage UI/UX validation. The results show that such a framework should be designed as a controlled, human-in-the-loop decision-support tool rather than a fully autonomous evaluator. The most important design factors were not the technical implementation itself, but the structure of the prompts, the specificity of the personas, the visibility of the reasoning process, and the ability for the designer to inspect and guide the simulation.

The final framework supports early-stage UI/UX validation by allowing synthetic users to process static UI states, reason through predefined tasks, and generate structured think-aloud outputs. Across the three prototypes, the framework improved from a basic screenshot-based reasoning loop into a more controlled persona-driven simulation approach. However, the results also show that persona adherence, behavioral realism, and trust remain partial rather than fully solved. The framework can therefore support early design reflection and identify possible usability friction, but its outputs still require expert interpretation and should not be treated as a replacement for human user testing.

4.3 Expert Evaluation of the Synthetic User Framework (RQ3)

The objective of RQ3 was to evaluate how the final AI-based synthetic user framework was perceived by the UI/UX expert in relation to early-stage UI/UX validation. The evaluation focused on perceived usefulness, insight quality, behavioral realism, trust, limitations, and requirements for practical adoption. The results are based on a written evaluation form completed by the single domain-expert after finishing the session, together with supporting comments from the session.

The main objective of RQ3 was to evaluate how the final AI-based synthetic user framework was perceived by the company expert, specifically regarding its applicability as a decision-support tool in early-stage UI/UX validation. Rather than focusing on quantitative performance metrics, this evaluation focused on capturing detailed, qualitative perspectives on the framework's practical viability within an established design workflow. To structure this analysis, the evaluation data came from the post-session evaluation form and was split across six different core dimensions: Perceived Usefulness, Insight Quality and Actionability, Behavioral Realism, Trust in Outputs, Identified Limitations, and Requirements for Practical Adoption. The following subsections present a detailed breakdown of the expert's feedback across the different dimensions, providing a comprehensive assessment of the framework's current capabilities and its necessary changes/improvements for future industry integration.

The expert perceived the framework as useful for early-stage UI/UX validation, particularly because recruiting real users can be difficult and expensive. The expert stated that LLM-based synthetic users should not be expected to replace real human testers. Instead, it was viewed as a complementary decision-support tool that can support early design reflection, identify potential usability issues, and guide further investigation.

4.3.1 Overall Perception of Usefulness

The UI/UX expert described the overall usefulness of the framework as high. The framework was considered particularly valuable in contexts where finding real users is difficult and expensive. In such situations, the expert perceived the tool as a useful alternative for early-stage validation.

The expert's response indicates that the framework could support faster design iteration by providing early feedback before more resource-intensive human testing is conducted.

At the same time, the usefulness was described in relation to a specific context. The perceived value was strongest where access to real users is limited. Therefore, the result should not be interpreted as showing that the framework can replace human user testing, but rather that it can provide practical support when traditional validation is difficult to perform.

4.3.2 Perceived Insight Quality and Actionability

The generated outputs were perceived as helpful to a great extent for identifying relevant usability issues and design considerations. The expert specifically noted that the framework appeared capable of identifying UI-related considerations such as layout, size, and visual hierarchy. The expert also observed that the synthetic user was able to navigate through a flow, identify the next action, and determine when the flow had ended.

This indicates that the framework was perceived as capable of supporting task-based UI/UX evaluation, especially for visual and interaction-related aspects of a design. The expert's comments suggest that the generated outputs were not limited to surface-level observations, but also reflected an understanding of user flow and task progression.

The expert also perceived the outputs as actionable. The generated insights were considered capable of supporting concrete design decisions and further investigation. In particular, the framework was seen as useful for early-stage design tasks and for refining UX flows and UI details. This suggests that the framework can contribute to practical design work by highlighting issues that designers can either address directly or investigate further through additional validation.

4.3.3 Perceived Behavioral Realism

The synthetic user's behavior and reasoning were perceived as realistic to a great extent when compared to real users. The expert stated that while the LLM should not be expected to replace real human testers, its behavior compared well to real users in the evaluated context.

A key factor contributing to this perception was that the personas and their personality traits appeared to be reflected in the synthetic user's reasoning and answers. The expert also observed that different personas behaved differently when completing the same task. This increased confidence that the persona design influenced the generated behavior in a meaningful way.

These results suggest that the framework achieved a degree of behavioral realism sufficient for early-stage design reflection. However, the expert's response also reinforces an important limitation: synthetic users should not be treated as equivalent to real human users. The realism was perceived as useful, but still bounded by the fact that the behavior is generated by an LLM rather than by an actual participant.

4.3.4 Trust in Generated Outputs

The expert reported trusting the generated outputs to a good extent. One factor that increased trust was that the expert compared the framework's results with previous real user testing and observed similar issues appearing. This comparison appeared to strengthen confidence in the framework, since the AI-generated outputs aligned with observations from human testing.

Another factor that increased trust was the framework's use of motivated answers and observations. The expert emphasized that explanations were important for building trust, since they made it possible to understand why the synthetic user reached a certain conclusion. This suggests that transparency and reasoning traces are central to the perceived credibility of AI-generated usability insights.

The expert also gained trust from seeing different personas behave differently on the same task. This indicated that the framework was not simply producing generic responses, but was able to reflect persona differences in its reasoning and behavior.

4.3.5 Identified Limitations and Obstacles

The expert identified several limitations in the current version of the framework. One limitation was that navigation through the UI flow was manual. The sequence of images reviewed by the LLM had to be defined manually, which slowed down the process. The expert also noted that this introduced a learning curve, although the learning curve was not perceived as steep.

Another limitation was that the framework used static images. The expert noted that real usability testing often uses working prototypes, which can better mimic both design and functionality. This means that the current framework is limited in how well it can represent interaction behavior compared to higher-fidelity prototypes.

A further limitation concerned personas with disabilities and accessibility needs. The expert noted that results involving these personas may be more challenging to evaluate without extensive experience working with those user groups. At the same time, the expert emphasized that accessibility-related design considerations are important and can have significant consequences.

4.3.6 Differences Between AI Models

The expert noticed differences between OpenAI and Claude outputs. OpenAI was described as more machine-like and concise, while Claude was described as more human in tone and more wordy. However, the expert stated that the content of the answers was similar to a high extent.

The expert preferred reading the Claude summary because it felt more human and realistic. This influenced the perception of the persona, since Claude's tone was seen as resembling a real human more closely. The expert also stated that this could increase trust in the framework as a whole.

4.3.7 Requirements for Practical Adoption

The expert considered the current version of the framework sufficient as an MVP. However, several improvements were suggested for future use.

The expert suggested that a dedicated user interface would improve the framework. Another suggested improvement was the ability to switch between LLMs directly in the application. The expert also suggested integrating persona and task creation into the tool, so that these steps would not need to be performed outside the application.

Finally, the expert suggested that the framework should provide information about its limitations before use. Tips and tricks for using the framework were also mentioned as potentially useful.

4.3.8 Summary of RQ3 Results

The RQ3 evaluation showed that the framework was perceived positively by the UI/UX expert. The expert described it as highly useful, especially in contexts where recruiting real users is difficult or expensive. The generated outputs were perceived as helpful for identifying UI-related considerations, navigating task flows, and supporting further design decisions.

The expert also perceived the synthetic user behavior as realistic to a great extent, while still emphasizing that synthetic users should not replace real human testers. Trust was increased by the presence of motivated answers, similarities to previous human-user testing results, and observable differences between personas.

The main limitations concerned manual navigation, reliance on static images, and the challenge of evaluating accessibility-related personas. For future adoption, the expert suggested a dedicated UI, integrated persona and task creation, model switching, and clearer information about tool limitations.

Chapter 5

Discussion

5.1 Overview of Findings

This chapter discusses the main findings of the thesis in relation to the research questions and the overall aim of exploring AI-based synthetic users for early-stage UI/UX validation. The results indicate that synthetic users can provide useful support during early design work, but that their value depends strongly on how the system is positioned and controlled.

Rather than showing that AI can replace human user testing, the findings point toward a more limited but practical role: supporting designers with early indications of possible usability issues before more resource-intensive validation is conducted. The developed framework was most useful when treated as a human-in-the-loop decision-support tool, where the designer remains responsible for interpreting the generated feedback.

A recurring finding across the study is that output quality is more important than automation level. The usefulness of the framework depended on whether the generated reasoning appeared understandable, relevant, and sufficiently realistic for early design reflection. This shaped both the final design of the framework and the interpretation of its practical value.

The following sections discuss these findings in more detail by addressing the shift from automation to decision support, the limits of behavioral realism, the role of personas, model dependency, and the requirements for practical adoption.

5.2 Interpretation of Results

5.2.1 From automation to Decision support

A central finding across the study is that the value of GenAI in early-stage UI/UX validation should not be understood primarily as full automation. The initial motivation for the framework was connected to reducing effort in traditional validation activities, such as preparing tests, recruiting participants, conducting sessions, and analyzing results. This was supported by the RQ1 findings, where activities such as synthetic user simulation, script drafting, transcription, prototype preparation, layout feedback, and behavioral flow analysis were identified as possible automation candidates.

However, RQ1 also showed that important parts of UX validation remain dependent on human judgment. The analysis identified human-in-the-loop boundaries

related to insight interpretation, subjective UX judgment, contextual understanding, and validation of AI-generated outputs. This means that automation alone is not sufficient if the generated feedback cannot be interpreted and verified by a designer.

This became central during RQ2. The design exploration compared alternatives with different levels of automation, where the more automated options offered better scalability and reduced manual effort, but also introduced concerns related to technical complexity, reliability, or incorrect flow inference. The selected MVP was therefore the controlled manual-loop approach using static screens. Although this reduced scalability, it provided greater transparency, controllability, and reliability for evaluating the generated outputs.

The prototype iterations reinforced this shift. Prototype 1 showed that the basic reasoning-action loop was technically feasible, but feasibility alone did not produce sufficiently useful or realistic outputs. Later prototypes therefore focused less on increasing automation and more on prompt structure, persona guidance, traceability, and the designer's ability to inspect the generated reasoning.

The final framework should therefore be understood as a decision-support tool rather than a fully autonomous evaluator. It supports early design reflection by generating possible usability friction and traceable think-aloud reasoning, while the designer remains responsible for interpreting, validating, and acting on the outputs. This is also consistent with the RQ3 findings, where the framework was perceived as useful for early exploration and rapid iteration, but not reliable as a standalone decision-maker.

The study suggests that GenAI's main role in this context is not to remove the human users from the validation process, but to support the designer with earlier and more structured feedback.

5.2.2 Behavioral Realism and the Limits of Synthetic Users

A central limitation identified during the study was the difficulty of making synthetic users behave like real users rather than rational task-solving agents. Across the prototype iterations, the AI-generated users were often more structured, consistent, and goal-oriented than human participants would typically be. This was especially visible in the earlier iterations, where the model often followed an optimal path through the interface and showed limited variation in interaction patterns.

This limitation is important because real users do not always interact with an interface in a logical or efficient way. In the RQ1 findings, human validation was described as valuable partly because users can misunderstand tasks, explore interfaces in unexpected ways, and reveal issues that designers did not anticipate. The synthetic users, in contrast, rarely showed this type of unpredictable behavior. They were less likely to hesitate, make mistakes, misunderstand instructions, or explore irrelevant paths unless such behavior was strongly encouraged by the prompt or persona design.

The prototype iterations improved this issue to some extent. Prototype 2 introduced more explicit think-aloud reasoning, which made the model's reasoning easier to inspect. Prototype 3 then introduced stronger persona definitions, which improved behavioral variation and made the synthetic users more distinguishable. However, these changes did not fully solve the realism problem. The generated behavior re-

mained more structured and logical than human behavior, and the model could still appear to reason more like an evaluator than a limited end user.

This suggests that behavioral realism in synthetic users should be understood as partial rather than complete. The framework can generate useful indications of possible usability friction, especially when the task, persona, and UI state are clearly defined. However, it cannot fully reproduce the uncertainty, irrationality, emotional reaction, or accidental behavior that may appear in real human user testing. For this reason, the outputs should be treated as early design signals rather than as direct evidence of how real users would behave.

The expert's responses indicate that trust depended on the outputs being interpretable, explainable, and comparable to known usability findings. Therefore, the framework's trustworthiness appears to depend not only on the quality of the generated issues, but also on how clearly the system explains its reasoning and how well the designer can evaluate the output.

The main implication is that synthetic users can support early-stage validation, but their realism has clear boundaries. They are useful for structured reflection on user flows and possible friction points, but they do not remove the need for human interpretation or later validation with real users.

5.2.3 The Role of Personas in Controlling AI Behavior

The results from RQ2 indicate that personas played an important role in controlling the behavior of the synthetic users. Across the prototype iterations, stronger persona definitions and more explicit behavioral constraints helped reduce generic assistant-like responses and made the synthetic users more distinguishable from each other.

However, the results also show that personas did not fully solve the problem of behavioral realism. Even when the personas were more detailed, the model could still drift toward coherent, helpful, or overly logical reasoning. This suggests that personas can guide LLM behavior, but they do not fully determine it.

The exploratory persona experiment in Prototype 3 used characters inspired by Winnie-the-Pooh because the characters have strong and clearly distinguishable personalities. The purpose was not to evaluate the fictional characters themselves, but to use them as exaggerated personality profiles, or edge cases, for testing whether the framework could produce visibly different behavior across personas. Since the characters differ strongly in traits such as confidence, anxiety, caution, curiosity, and impulsiveness, they provided a useful way to stress-test whether persona definitions could influence the synthetic user's reasoning and interaction behavior.

The results suggest that these recognizable personas provided stronger contextual cues for the model. In one transcript, the synthetic user based on Winnie-the-Pooh referred to Christopher Robin, even though this detail was not part of the persona or the immediate UI task. This indicates that the model drew on prior knowledge associated with the character, rather than only following the explicit task instructions. Such prior knowledge may help the model maintain a more coherent role, but it also makes the source of the generated behavior harder to verify.

At the same time, this result should be interpreted carefully. Because the internal reasoning of the LLM is not directly observable, it is difficult to determine whether the observed behavior was caused by the explicit persona design, the model's prior

knowledge of the character archetypes, or a combination of both. This represents a black-box limitation in the framework.

The findings suggest that persona design is central to synthetic-user simulation, but that persona fidelity remains partial. Personas can improve behavioral differentiation and reduce generic outputs, but they cannot guarantee that the generated behavior fully reflects the intended user profile.

5.2.4 Model Dependency

The RQ3 evaluation also showed that the framework is partly dependent on the underlying LLM. The expert perceived OpenAI and Claude as producing largely similar content, but with different presentation styles. OpenAI was described as more concise and machine-like, while Claude was perceived as more human in tone and more wordy.

This difference affected the perceived realism of the synthetic persona. The expert preferred the Claude summary because it felt more human and realistic, and stated that this could increase trust in the framework. This indicates that model dependency is not only related to the content of the generated insights, but also to tone, wording, and presentation style.

However, the more human and detailed tone of Claude also introduced a practical trade-off. Longer and more elaborate responses may increase the cognitive load for the designer, since the generated logs require more time and effort to read, compare, and interpret. In contrast, OpenAI's more concise outputs may be easier to process quickly, even if they are perceived as less human in tone. This suggests that model selection involves a balance between perceived realism and practical readability.

This is important because the framework relies on the designer's interpretation of qualitative outputs. If two models identify similar issues but present them differently, the perceived trustworthiness of the framework may still differ. Therefore, model selection should be considered part of the framework design, especially when the goal is to simulate human-like reasoning rather than only generate concise usability observations.

At the same time, the comparison between models was limited. The Claude run was used as a supplementary comparison rather than a formal benchmark. Therefore, the results should be interpreted as an indication that model choice may affect perceived realism and trust, rather than as evidence that one model is generally better suited for synthetic user testing.

5.2.5 Requirements for Practical Adoption

The RQ3 results indicate that the current framework was perceived as sufficient as an MVP, but that several improvements would be needed before regular workflow adoption. The expert emphasized the need for a dedicated user interface, integrated persona and task creation, and the ability to switch between LLMs directly within the application.

These requirements suggest that the practical barrier is not only whether the synthetic users can generate useful outputs, but also how easily the framework can be used within an existing design workflow. In its current form, the process still

requires manual navigation, preparation of personas and tasks outside the tool, and interpretation of static image sequences. This creates friction for repeated use.

The expert also suggested that the framework should communicate its limitations more clearly before use. This is important because the framework produces outputs that may appear confident and human-like, even when they are based on simulation rather than real user behavior. Clear limitation guidance could help designers interpret the outputs more critically and avoid over-reliance on AI-generated conclusions.

Therefore, practical adoption depends on both usability improvements and transparency. A more mature version of the framework would need to reduce setup effort, support smoother interaction with UI flows, and make the boundaries of AI-generated validation explicit to the designer.

5.3 Implications

The findings of this study have implications for both UI/UX practice and future research on AI-based usability validation. The results suggest that AI-based synthetic users are most valuable when used as an early-stage decision-support tool rather than as a replacement for human user testing.

The framework can support designers by enabling lightweight usability-oriented evaluations before a design is mature enough to justify full human testing. However, the results also show that the usefulness of such systems depends strongly on the realism of the generated behavior, the clarity of the generated reasoning, and the designer's ability to critically interpret the outputs.

5.3.1 Implication for Practice

For practitioners, the main implication is that AI-based synthetic users can lower the threshold for conducting early validation. In current UI/UX workflows, validation with real users is often limited by time, cost, recruitment effort, and project constraints. The framework explored in this study indicates that synthetic users may help designers identify potential usability issues earlier in the design process, especially during exploratory phases where rapid iteration is more important than final validation.

The framework should therefore be used to support early exploration, compare alternative design directions, and reveal possible cognitive friction before investing resources in human testing or implementation. In this role, the AI can help designers ask better questions about a design and highlight areas that may require further investigation. This makes the approach especially relevant for static prototypes and early Figma-based designs, where traditional testing may not yet be feasible or cost-effective.

However, the findings also imply that AI-generated validation should not be treated as equivalent to human user testing. Synthetic users do not have genuine emotions, lived experience, or unpredictable human behavior. They may also produce confident but incorrect interpretations of an interface. Therefore, AI-generated findings should be interpreted as preliminary indicators rather than verified usability

problems. Designers should use the outputs as input for reflection, discussion, and prioritization, but not as the sole basis for final design decisions.

Another practical implication is that successful adoption depends on workflow integration. The evaluation showed that the current framework was perceived as useful as an MVP, but that practical use would require a dedicated interface, easier persona and task creation, clearer communication of limitations, and smoother support for switching between models. This means that the value of synthetic user testing is not determined only by the quality of the AI model, but also by how easily the tool fits into existing design practices.

Finally, the study suggests that persona and prompt design should be treated as central parts of the UX validation process. The quality of the generated insights was closely connected to how the synthetic users were defined and constrained. Practitioners who use similar approaches should therefore avoid generic AI prompts and instead define clear task goals, user characteristics, behavioral constraints, and expected output formats.

5.3.2 Implication for Research

For research, the findings suggest that the evaluation of AI-based UX validation tools should move beyond automation efficiency alone. Much of the value of the framework was not found in fully automating usability testing, but in generating outputs that were realistic, interpretable, and useful for designers. This indicates that future research should place stronger emphasis on behavioral realism, insight quality, trust, and human-AI collaboration.

The study also highlights the importance of investigating the role of personas in synthetic user testing. The iterative development process showed that persona definitions and prompt structures influenced how human-like, useful, and action-oriented the outputs became. Future research should therefore examine how different persona construction methods affect synthetic user behavior, and whether psychometric or empirically grounded persona models can improve consistency and realism across different tasks and interfaces.

Another implication is that model dependency should be treated as a research concern. The comparison between OpenAI and Claude indicated that different models may produce similar types of usability observations while differing in tone, detail, readability, and perceived human-likeness. This suggests that the choice of LLM can influence designer trust and interpretation, even when the underlying usability issues are similar. Future studies should therefore compare multiple models more systematically and investigate how model-specific output styles affect perceived usefulness and trust.

The findings further imply that human-in-the-loop evaluation remains necessary when applying AI to UX validation. Rather than removing the designer from the process, the framework depends on the designer to interpret the generated reasoning, judge its relevance, and decide whether an issue should influence the design. This supports a research direction where AI systems are studied as collaborative design-support tools rather than autonomous evaluators.

Finally, the study contributes to research by showing that early-stage static UI validation is a meaningful context for synthetic user testing. Existing work often

focuses on implemented systems, HTML environments, or fully autonomous agents. This thesis instead demonstrates the relevance of controlled, screenshot-based, task-oriented validation for early design phases. Future research can build on this by evaluating the approach across more design cases, more participants, and more diverse user groups to assess how transferable the findings are beyond the single case context.

5.4 Reflection on Method and Design Choices

A central design choice in this thesis was to prioritize a controlled, human-in-the-loop framework instead of a highly automated solution. Although more automated alternatives were considered, the final approach was selected because it offered a better balance between feasibility, transparency, behavioral realism, and evaluation quality.

This decision was based on the insight that automation alone does not guarantee useful results. During development, it became clear that the value of the framework depended mainly on whether the AI-generated outputs were realistic, understandable, and useful for design decisions. A more automated system would have been less valuable if it produced robotic or difficult-to-interpret feedback.

The stepwise interaction approach, where the synthetic user only sees one UI state at a time, helped make the evaluation more similar to a real task-based usability test. It also made the AI's reasoning easier to follow and allowed the designer to remain in control of the process. This supported the goal of positioning the framework as a decision-support tool rather than a replacement for human user testing.

However, this choice also introduced trade-offs. The controlled approach improved transparency and realism, but reduced scalability and required more manual effort from the designer. The evaluation was also based on qualitative expert feedback, which means that the findings should be seen as indicative rather than generalizable.

Overall, the methodological choices shaped the framework toward depth rather than scale. This was appropriate for the exploratory aim of the study, where the main goal was to investigate whether AI-based synthetic users can support early-stage UI/UX validation in a realistic, trustworthy, and practically useful way.

5.5 Limitations

While the Design Science Research (DSR) methodology incorporated continuous evaluation to mitigate risks during the development phases (as detailed in 3.1), several inherent limitations remain within the finalized framework. These limitations are categorized into three different section, development limitations, AI limitations, and evaluation limitations.

5.5.1 Development Limitations

The artifact developed in this thesis serves as a proof-of-concept design-support framework rather than a production-ready software tool. Because the system's de-

sign intentionally prioritized behavioral realism and transparent reasoning over full automation (Option B), the framework currently lacks autonomous scalability. It requires continuous human-in-the-loop orchestration to advance UI states and interpret findings. Furthermore, technical debt remains regarding system integration, were the framework currently processes static, manually captured screenshots. A fully realized production version would require direct API integration into design environments, such as Figma, to facilitate seamless testing workflows.

Because the framework relied on static screenshots, it could not fully capture dynamic interaction behavior such as scrolling, hover states, animations, form input, loading states, validation errors, or timing-related usability issues. This limits the types of usability problems the framework can detect compared to testing an interactive prototype or implemented system.

An important limitation to take into consideration is that the developed framework is one specific approach to persona generation. While the chosen architecture, combining psychometric parameters with rigid prompt constraints provides a scientifically grounded foundation for the synthetic users, it represents only one possible instantiation within a very broad design space. This thesis does not claim that this specific configuration is the universally optimal method for human behavioral simulation. Therefore, the chosen framework should be viewed as a functional proof-of-concept that successfully addresses the research questions, rather than the definitive or most optimal solution for AI-based UX validation.

5.5.2 AI Limitations

A fundamental limitation of utilizing Large Language Models (LLMs) for user simulation is the inherent "rationality bias" of the models. Despite the implementation of strict BFI-2 psychometric parameters and negative constraints, models like GPT and Claude are still optimized to be helpful, logical, and structured. Where real human users are frequently erratic, misunderstand instructions, or explore interfaces irrationally. While the synthetic users successfully simulated cognitive friction, it still remained more systematic and goal-oriented than true human subjects at times, failing to fully capture authentic human unpredictability. Finally, the framework's reliability is entirely dependent on the actual models (e.g., GPT-5.4, Claude Opus 4.6). Because LLMs are "black boxes," it is impossible to definitively prove why the model made a specific choice, whether it genuinely adhered to the assigned persona or simply defaulted to its internal training biases. Furthermore, as these models are continuously updated by their providers, the reproducibility of the specific outputs observed in this study cannot be guaranteed long-term.

5.5.3 Validation and Evaluation Limitations

The evaluation of this framework was conducted as a qualitative case study within a single organization (Strawberry Planet), utilizing a single UI/UX expert. While this allowed for continuous, iterative feedback aligned with DSR methodologies, it severely limits the external validity and transferability of the findings. The perceived usefulness and behavioral realism of the framework are highly contextual to Strawberry Planet's specific design structure and workflow. Furthermore, the evaluation

relies on subjective, qualitative metrics. "Behavioral realism" and "insight quality" were assessed via expert perception rather than objective benchmarking.

A further limitation is that the framework was not evaluated through a direct controlled comparison with real users performing the same tasks on the same interface. As a result, the study cannot determine whether the synthetic users identified the same usability issues as human participants, missed critical issues, or introduced false positives. The comparison to human user testing is therefore based on expert judgment and prior experience rather than empirical equivalence.

The study was also limited by the number and diversity of synthetic personas tested. Although the framework used structured persona definitions, these personas cannot fully represent the diversity of real user populations. Factors such as accessibility needs, cultural background, prior domain knowledge, motivation, language ability, and situational context may strongly influence usability behavior, but were only partially captured in the synthetic user setup.

The evaluation was conducted using a limited set of UI designs and task scenarios. Therefore, the results may be influenced by the specific interface structure, visual style, and task complexity used in the study. The framework may perform differently for other types of systems, such as data-heavy dashboards, mobile applications, e-commerce flows, accessibility-focused interfaces, or highly domain-specific professional tools.

Finally, although the framework is motivated by reducing the cost and effort of early-stage validation, this study did not quantitatively measure time savings, cost reduction, or productivity improvements. The practical efficiency benefits of the framework therefore remain assumed or perceived rather than empirically demonstrated.

5.6 Validity and Reliability

5.6.1 Reliability / Dependability

AI models such as OpenAI's ChatGPT are constantly updated and refined, making model version drift a limitation for consistent results. Constant updates to LLMs can therefore lead to inconsistency and reproducibility limitations.

Furthermore, the development of the framework relied on a specific version of ChatGPT available at the time of the study. This introduces a potential model-specific bias, where the observed behavior and results may be dependent on the characteristics of that particular model version rather than generalizable across LLMs.

LLMs can also struggle to maintain a consistent persona throughout a long session, as they may become more sensitive to immediate visual stimuli than to predefined persona constraints. This introduces variability in behavior across simulation runs.

Finally, LLM outputs are inherently sensitive to prompt formulation and input structure. Small variations in prompts or context can lead to different outputs, which introduces further challenges for reproducibility and consistency of results. Even the same prompt and inputs can give variations in the results

5.6.2 Internal Validity / Credibility

Internal validity concerns whether the study accurately evaluates the intended concepts, namely the behavioral realism, insight quality, and practical usefulness of AI-based synthetic users in early-stage UI/UX validation.

Synthetic users lack genuine human intuition, emotional reactions, and the limited ability to interpret subjective and context-dependent visual elements, which is a critical part of high-fidelity UX validation.

AI-simulated users are often perceived as overly rational compared to real users, while real users may behave unpredictably, misunderstand instructions, or explore interfaces irrationally. Simulating such behavior is difficult for current AI systems. This could generate a lot of false truths.

Additionally, AI-generated outputs may contain false positives or false negatives, where the system identifies non-existent usability issues or fails to detect critical ones. This affects the credibility of the generated insights.

When working with usability issues, there is rarely one "ground truth," which limits the ability to objectively validate the framework's output quality. Most results from usability testing are subjective to the specific user or designer, making objectively identified usability issues difficult to prove.

A specific credibility threat in this study is that the same UI/UX expert was involved across several stages of the research process. The expert participated in RQ1 as an interviewee, contributed to RQ2 as a domain informant and formative evaluator during the iterative refinement of the framework, and acted as the sole evaluator in RQ3. This strengthened the practical relevance of the artifact, since the framework was closely aligned with an authentic Strawberry Planet workflow. However, it also introduces a risk of confirmation bias, as the expert may have evaluated the final framework more positively due to prior involvement in its design. No additional evaluator was included due to the exploratory scope of the study, limited access to comparable UI/UX practitioners during the project period, and the intention to evaluate the framework within the specific workflow it was designed to support.

5.6.3 External Validity / Transferability

The study is conducted in collaboration with Strawberry Planet, which introduces a risk of confirmation bias where the framework is optimized toward their specific needs and design style rather than universal usability standards.

This study relies on the evaluation of a single UI/UX expert, which limits external validity. The availability of the expert also constrained the frequency and depth of feedback during development and evaluation.

The results should therefore be interpreted as indicative of practical usefulness rather than statistically generalizable conclusions. Results should not be interpreted as universally applicable across industries or design cultures.

Testing user validation on prototypes limits the number of usability aspects possible to test. Aspects such as latency, bugs, input/output devices, and error codes cannot be tested on a Figma prototype. This also leads to limited coverage of accessibility users such as screen reader users or keyboard-only users.

5.6.4 Measures Taken to Strengthen Trustworthiness

To address the identified threats to validity and reliability, several measures were implemented throughout the study to strengthen the trustworthiness of the results.

5.6.4.1 Control and Consistency of AI Behavior

During the design and development phase, prompts and task definitions were iteratively refined based on expert feedback to improve the quality, realism, and usefulness of the generated outputs. While this iterative process introduces variability, it is consistent with the Design Science Research approach, where continuous refinement is required to develop a functional artifact.

To ensure comparability and reduce variability during the evaluation phase, a fixed set of prompts, personas, and task scenarios was used. In addition, a controlled stepwise interaction approach was implemented, where the AI was exposed to one UI state at a time. This reduced full-context bias and improved the realism of simulated interactions.

5.6.4.2 Transparency and Interpretability

All AI generated outputs were clearly labeled to avoid automation bias. Furthermore the framework generated "think-aloud" reasoning logs, both for the usability insights and the ability to trace the AI conclusions. This increased transparency and enabled critical evaluation of the AI's reasoning process instead of relying on black-box outputs

5.6.4.3 Human-in-the-loop evaluation and iterative refinement

The development process involved continuous expert feedback where the UI/UX expert participated both as a domain informant and evaluator. Iterative refinement cycles were conducted to improve behavioral realism, quality and practical usefulness of the design.

5.6.4.4 Researcher Bias and Reflexivity

The Design Science Research approach involves active researcher participation in the design, development, and evaluation of the artifact. In this study, the researchers were involved in defining personas, designing prompts, and interpreting AI-generated outputs, which introduces a risk of researcher bias, particularly confirmation bias.

To mitigate this, continuous involvement of a UI/UX expert provided an external perspective during both development and evaluation. In addition, predefined evaluation criteria was defined prior to evaluation.

5.6.4.5 Realistic evaluation context

Test scenarios were derived from actual design workflows withing the case organization. This ensures that the evaluation reflects real-world use cases. In addition, expert assessments were complemented by comparisons to earlier real user validation

testing experiences. These practical references could help guide the design and evaluate the quality of AI-generated insights, which were used to structure the analysis and reduce subjective interpretation.

While researcher influence cannot be fully eliminated, these measures help ensure that the evaluation remains grounded in professional UX practice rather than solely researcher assumptions.

5.6.4.6 Variation and Robustness of Simulation

To improve robustness and reduce model-specific bias, the framework was evaluated across multiple LLMs during the evaluation phase (RQ3). While the development phase (RQ2) was conducted using ChatGPT, the same personas, prompts, and task scenarios were later tested using an additional LLM (Claude).

This allowed for comparison of behavioral patterns, reasoning consistency, and variation in generated insights across models while keeping the experimental setup constant. Models such as Gemini were excluded during earlier stages due to feasibility constraints identified during initial exploration 4.2.1.

Although this multi-model evaluation does not eliminate model-specific bias, it provides an indication of how sensitive the framework is to different underlying LLMs and strengthens the robustness of the findings.

5.6.4.7 Validation of expert-based evaluation

While expert evaluation introduces subjectivity, it reflects common practice in professional UX workflows. The use of transparent outputs, structured evaluation criteria, and iterative feedback helped ensure that assessments were grounded, consistent, and aligned with real-world usability evaluation practices.

5.7 Ethical, Societal, and Sustainability Considerations

The use of AI-based synthetic users in early-stage UI/UX validation introduces several ethical, societal, and sustainability considerations. Since the framework is intended to simulate user behavior and generate usability insights, it is important to consider how such outputs should be interpreted and used in practice. The framework developed in this thesis was not designed to replace human users, but to support designers during early design iterations. Therefore, the ethical use of the framework depends on maintaining a human-in-the-loop approach, where the designer remains responsible for interpreting, questioning, and validating the AI-generated outputs.

5.7.1 Ethical and Societal Considerations

The main ethical risk is over-reliance on AI-generated feedback. Synthetic users can produce convincing explanations, but they do not have real experiences, emotions, needs, or accessibility limitations. Therefore, their feedback should be treated as indicative rather than as confirmed usability evidence.

Responsibility also remains with the human designer. If the AI identifies an incorrect issue, or fails to identify an important one, the output still needs to be interpreted and judged by a human. This is why the human-in-the-loop approach is important in this thesis. This aligns with the "Human-AI Interaction Guidelines" [3], suggesting that AI should support rather than replace human intuition.

There is also a societal risk if organizations use synthetic users only because they are cheaper or faster than real user testing. This could reduce the involvement of real users in design processes. However, when used carefully, synthetic users may also make early validation more accessible for smaller companies with limited time and resources.

5.7.2 Sustainability Considerations

The framework also has sustainability implications. From an economic perspective, early-stage validation can support more sustainable software development by identifying usability issues before implementation. If design problems are found earlier, fewer resources may be spent on rework later in the development process. This aligns with the shift-left principle discussed in this thesis, where validation is moved earlier to reduce the cost and impact of late changes.

At the same time, AI-based validation introduces environmental costs. Large Language Models require computational resources, which consume energy and may contribute to environmental impact when used at scale [11]. In this thesis, the number of tests was limited and the direct environmental impact was therefore small. However, if similar systems were adopted more broadly in industry, the accumulated compute usage could become more significant.

There is therefore a trade-off between reducing resources spent on late-stage redesign and increasing the use of computationally intensive AI systems. Whether the overall effect is positive or negative depends on how the framework is used. If synthetic users are used selectively to support early design decisions and reduce unnecessary rework, the approach may contribute to more efficient development. However, if large numbers of AI simulations are generated without clear purpose, the sustainability benefit may be reduced.

Future versions of the framework should therefore consider more resource-efficient usage. This could include limiting the number of simulation runs, using smaller or local models when possible, reusing prompts and evaluation structures, and applying AI only when it adds clear value to the design process. In this way, AI-based UX validation can be used in a more responsible and sustainable manner.

Chapter 6

Conclusions and Future Work

6.1 Conclusion

This chapter summarizes the main findings of the thesis and answers the research questions formulated in Chapter 1.1. The study's practical and scientific contributions to the research field are presented, followed by recommendations for future research and technical development.

6.2 Answer to Research Questions

6.2.1 RQ1 Conclusion

How is early-stage UI/UX validation currently conducted in practice, and which activities have the greatest potential for GenAI-based automation? The initial phases of the study confirmed that current methods for early-stage UX validation are highly resource and time-intensive. Due to the high costs and logistical challenges of recruiting human test subjects, validation is often selective and omitted during the conceptual phase, leading to usability issues being discovered late in the development cycle. The findings demonstrate that Generative AI has significant potential to serve as a decision-support tool in these early stages. However, the study concludes that AI should not operate entirely autonomously, therefore having a "Human-in-the-Loop" approach is essential to filter insights, prevent automation bias, and maintain designer trust.

6.2.2 RQ2 Conclusion

How can an AI-based synthetic user framework utilizing personas be designed and refined to effectively support early-stage UI/UX validation? To address the problems identified from previous automated tools, a framework was developed based on controlled interaction and structured prompt architectures. A key finding from the exploratory design phase was that the architecture must prioritize output quality and behavioral transparency over full autonomy. By selecting a controlled manual-loop approach operating on static screens (option B), where the designer steps through the interface alongside the AI, hallucinations were minimized and trust increased due to improved transparency and control. The synthetic agents were modeled by combining the "Intent/Domain/Demand" taxonomy with the BFI-2

psychometric framework. This forced the Large Language Model (LLM) to act based on specific human limitation, emotional prioritization, and technological uncertainty.

6.2.3 RQ3 Conclusion

How is the AI-based synthetic user framework perceived by a UI/UX expert in terms of usefulness, behavioral realism, trust, and practical applicability for early-stage UI/UX validation? The evaluation, conducted with the domain expert from Strawberry Planet, demonstrated that the framework can successfully identify basic usability issues and provide structured reasoning regarding cognitive friction. Through the enforced "Think-Aloud" protocol and negative constraints, the framework succeeded in changing the AI's standardized expert tone, generating high behavioral realism in specific interactions. However, inherent limitations remain, for example, the AI still exhibits a fundamental "rationality bias" and can still sometimes lack the true unpredictability of a human user. The framework is therefore deemed capable of generating highly relevant and actionable insights, but with the strict caveat that synthetic testers cannot and should not replace verification with real human user in the final stages of product development. But does still require validation with independent evaluators before broader conclusions can be drawn.

6.3 Overall conclusion

The conclusion of this study is that AI-based synthetic users offer a promising method for shifting UX validation significantly earlier in the software development lifecycle. By combining psychometric prompt design with human orchestration, the framework shows potential to mitigate the algorithmic aversion that often characterizes fully automated testing tools. The result is a decision-support system that can help build trust with designers and identifies cognitive friction before code has been written, promoting a more resource-efficient and sustainable software engineering process.

6.4 Contributions

6.4.1 Practical Contributions

The main practical contribution of this study is the development of a functional proof-of-concept framework, tailored specifically for early-stage design workflows at Strawberry Planet. Looking past the artifact itself, this thesis provides concrete, actionable guidelines for UI/UX practitioners on how to effectively integrate GenAI into their validation processes. Furthermore, the study offers explicit insights into the boundaries of current AI models, clearly outlining the scenarios where AI shows high potential as a decision-support tool (such as evaluating static visual hierarchies and navigation logic) versus where it currently falls short (such as evaluating dynamic micro-interactions and authentic human unpredictability).

6.4.2 Scientific Contributions

Scientifically, this thesis contributes to the emerging Human-Computer Interaction (HCI) research domain surrounding synthetic users and early-stage UX validation. The study empirically demonstrates that the primary challenge of utilizing Large Language Models (LLMs) for usability testing is no longer technical automation, such as enabling an AI to autonomously click through an interface, but rather achieving authentic behavioral realism. By demonstrating that parameter-free optimization and psychometric prompting (via the BFI-2 framework) can help mitigate an AI's inherent "rationality bias," this research lays foundational work for future academic studies on AI-driven behavioral simulation and Human-AI collaboration.

6.5 Future Work

Based on the framework's current limitations and the insights gained from the expert evaluation, several important areas for future research and development were found. These can be divided into three primary categories:

Technical Development and Integration The expert evaluation confirmed that the current framework functions successfully as a Minimum Viable Product (MVP). A natural next step for technical development would be to transition this MVP into a fully standalone application with a dedicated user interface (UI), or integrate it directly into a professional design tool like Figma. To reduce setup effort and enable a smoother operations, this future application should feature an integrated environment for creating personas and configuring tasks, eliminating the need for designers to manage prompts externally. Furthermore, the tool should include native functionality to seamlessly switch between different LLMs directly within the application, allowing designers to easily cross-verify usability insights across multiple models.

AI and Behavioral Modeling Although the psychometric approach significantly improved behavioral fidelity and persona consistency, future research should aim to explore even more robust persona designs to completely overcome the AI's "rationality bias" and simulate human-like unpredictability. Building upon the exploratory experiments conducted in this study, one promising path would be to investigate the trade-offs between utilizing standardized psychometric frameworks (such as BFI-2) and recognizable cultural or fictional personas. While the BFI-2 framework provides a scientifically grounded baseline, exploratory tests using well-known fictional characters (e.g., Winnie-the-Pooh) indicated that recognizable archetypes may provide stronger contextual cues, showing more consistent and strongly differentiated cognitive friction from the underlying LLM training. Future studies should investigate whether these recognizable profiles reduce the model's inherent rationality bias more effectively than abstract parameters. Importantly, this research must also address the "black box" limitation to determine whether the observed behavior comes from genuine persona adherence or simply the model's pre-trained knowledge of the character. Beyond cognitive and personality-driven variations, an equally important future

addition to persona modeling is the inclusion of accessibility-focused synthetic users (e.g., simulating color blindness, motor impairments, or reliance on screen readers) to broaden the inclusiveness of automated early-stage testing.

Evaluation, Trust, and Quantitative Validation Finally, the qualitative results from this study must be supplemented with rigorous quantitative evaluations and broader contextual testing. Because this framework was evaluated within a specific organizational context and a limited set of UI designs, improving its external validity is a must. Future research must evaluate the framework with more experts and organizations, and systematically test it across a wider variety of UI archetypes, such as data-heavy dashboards, mobile-first applications, or conversion-driven e-commerce flows. Expanding the evaluation to include these diverse environments will help clarify the framework's generalizability and identify which specific types of digital interfaces are most suitable for AI-driven early-stage validation versus those that strictly require human oversight.

Alongside expanding the testing environments, controlled empirical testing is necessary to investigate how the framework can practically reduce time, costs, and participant recruitment needs, while systematically measuring the occurrence of false positives against baseline human data. Furthermore, based directly on expert feedback from this study, an important component of building designer trust involves transparent onboarding. Future iterations of the tool must explicitly communicate the AI's limitations upfront and provide functional instructions on how to operate and fine-tune the tool to calibrate user expectations before a test begins. These rigorous evaluations and trust-building measures will clarify how AI-based validation can be effectively integrated into existing UX workflows over time.

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Appendix A

Supplemental Information

A.1 Image recognition results

A.1.1 Claude

Image Element Identification Summary

- Source log: 2026-04-15T13-39-00-242Z-image-elements.json
- Finished at: 2026-04-15T13:39:00.241Z
- Provider: anthropic
- Design name: Bad_UI_ex
- Image name: AirBnB.png
- Model: claude-opus-4-6
- Model version: claude-opus-4-6
- Element count: 34

#	Element	Type	Purpose
1	Airbnb logo	Logo/link	Brand logo that likely links to the homepage
2	Boenden tab	Navigation tab	Switch to the accommodations/stays search mode (currently active)
3	Upplevelser tab	Navigation tab	Switch to the experiences search mode, marked with 'NYTT' (new) badge
4	Tjänster tab	Navigation tab	Switch to the services search mode, marked with 'NYTT' (new) badge
5	Bli en värd	Text link	Navigate to become a host on Airbnb
6	Globe icon	Icon button	Change language or region settings
7	Hamburger menu	Icon button	Open user menu or navigation options
8	Var (Where) search field	Text input	Enter a destination to search for accommodations
9	När (When) search field	Text input	Select check-in and check-out dates
10	Vem (Who) search field	Text input	Specify the number of guests
11	Search button	Button	Execute the search with the specified criteria (red circular button with magnifying glass icon)
12	Populära boenden i Gotland heading	Heading with arrow link	Section title for popular accommodations in Gotland, with link to see more
13	Gotland carousel previous button	Icon button	Navigate to previous listings in the Gotland carousel (appears disabled)
14	Gotland carousel next button	Icon button	Navigate to next listings in the Gotland carousel
15	Stuga i Burgsvik listing card	Card	Display a guest favorite cabin listing in Burgsvik with dates, price (1,059 kr SEK), and rating (5.0)
16	Boende i Gotland listing card	Card	Display a guest favorite accommodation in Gotland with dates, price (2,118 kr SEK), and rating (4.99)
17	Stuga i Slite listing card	Card	Display a guest favorite cabin in Slite with dates, price (836 kr SEK), and rating (4.86)
18	Stuga i Fårö listing card	Card	Display a guest favorite cabin in Fårö with dates, price (1,648 kr SEK), and rating (5.0)
19	Stuga i Gotland listing card	Card	Display a guest favorite cabin in Gotland with dates, price (1,177 kr SEK), and rating (4.99)
20	Ägarlägenhet i Innerstaden listing card	Card	Display a guest favorite owner apartment in Innerstaden with dates, price (2,236 kr SEK), and rating (4.91)
21	Stuga i Gotland (second) listing card	Card	Display a guest favorite cabin in Gotland with dates, price (2,090 kr SEK), and rating (4.75)
22	Gästfavorit badge (multiple)	Badge/label	Indicate that the listing is a guest favorite
23	Heart/favorite icon (multiple)	Icon button	Save or wishlist a listing
24	Utvalda hotell i London heading	Heading with arrow link	Section title for selected hotels in London, with link to see more
25	London section subtitle	Text	Describes the section as a selection of independent and hand-picked hotels
26	London carousel previous button	Icon button	Navigate to previous hotel listings in the London carousel (appears disabled)
27	London carousel next button	Icon button	Navigate to next hotel listings in the London carousel

28	Regency House Hotel listing card	Card	Display a hotel listing for Regency House Hotel in London
29	Rockwell East listing card	Card	Display a hotel listing for Rockwell East in London
30	Grange Buckingham listing card	Card	Display a hotel listing for Grange Buckingham in London
31	Roomzzz Aparthotel London Stratford listing card	Card	Display a hotel listing for Roomzzz Aparthotel London Stratford
32	Hotel Saint listing card	Card	Display a hotel listing for Hotel Saint in London
33	Mercure London Hyde Park Hotel listing card	Card	Display a hotel listing for Mercure London Hyde Park Hotel
34	Grange Clarendon listing card	Card	Display a hotel listing for Grange Clarendon in London

A.1.2 Gemini

Image Element Identification Summary

- Source log: 2026-04-15T19-01-52-473Z-image-elements.json
- Finished at: 2026-04-15T19:01:52.472Z
- Provider: gemini
- Design name: Bad_UI_ex
- Image name: AirBnB.png
- Model: gemini-3.1-pro-preview
- Model version: gemini-3.1-pro-preview
- Element count: 18

#	Element	Type	Purpose
1	Airbnb Logo	Image/Logo	Navigates to the homepage.
2	Boenden Tab	Navigation Tab	Filters search for accommodations.
3	Upplevelser Tab	Navigation Tab	Filters search for experiences.
4	Tjänster Tab	Navigation Tab	Filters search for services.
5	Bli en värd Button	Button	Link to become a host.
6	Language Selection Icon	Icon Button	Opens language and currency settings.
7	User Menu Icon	Icon Button	Opens user account menu.
8	Search Destination Input	Text Input	Input field for destination search.
9	Search Dates Input	Text Input	Input field for check-in and check-out dates.
10	Search Guests Input	Text Input	Input field for number of guests.
11	Search Button	Button	Executes the search.
12	Section Title: Populära boenden i Gotland	Heading	Title for the popular accommodations section.
13	Listing Card (Stuga i Burgsvik)	Card	Displays details for a specific accommodation listing.
14	Listing Card (Boende i Gotland)	Card	Displays details for a specific accommodation listing.
15	Listing Card (Stuga i Slite)	Card	Displays details for a specific accommodation listing.
16	Section Title: Utvalda hotell i London	Heading	Title for the selected hotels section.
17	Listing Card (Regency House Hotel)	Card	Displays details for a specific hotel listing.
18	Listing Card (Rockwell East)	Card	Displays details for a specific hotel listing.

A.1.3 GPT

Image Element Identification Summary

- Source log: 2026-04-15T13-37-07-480Z-image-elements.json
- Finished at: 2026-04-15T13:37:07.479Z
- Provider: openai
- Design name: Bad_UI_ex
- Image name: AirBnB.png
- Model: gpt-5.4
- Model version: gpt-5.4-2026-03-05
- Element count: 30

#	Element	Type	Purpose
1	Airbnb logo	logo	Identifies the Airbnb brand and typically links back to the home page.
2	Boenden tab	navigation tab	Switches to the stays browsing mode for accommodations.
3	Upplevelser tab	navigation tab	Lets users browse experience-based activities.
4	Tjänster tab	navigation tab	Lets users browse services offered on the platform.
5	Bli en värd	link	Takes users to hosting information or host signup flow.
6	Globe button	icon button	Opens language or regional settings.
7	Menu/profile button	icon button	Opens the account and site navigation menu.
8	Var field	text input	Lets users enter a destination to search for places to stay.
9	När field	date input	Lets users choose check-in and check-out dates.
10	Vem field	guest input	Lets users specify the number of guests.
11	Search button	button	Runs the accommodation search using the selected destination, dates, and guests.
12	Populära boenden i Gotland	heading	Labels the carousel of popular stays in Gotland.
13	Gotland section next button	icon button	Moves the Gotland listings carousel forward.
14	Stuga i Burgsvik card	card	Shows a specific Gotland listing with photo, dates, price, and rating for selection.
15	Boende i Gotland card	card	Shows a specific Gotland listing with photo, dates, price, and rating for selection.
16	Stuga i Slite card	card	Shows a specific Gotland listing with photo, dates, price, and rating for selection.
17	Stuga i Fårö card	card	Shows a specific Gotland listing with photo, dates, price, and rating for selection.
18	Stuga i Gotland card 1	card	Shows a specific Gotland listing with photo, dates, price, and rating for selection.
19	Ägarlägenhet i Innerstaden card	card	Shows a specific Gotland-area listing with photo, dates, price, and rating for selection.
20	Stuga i Gotland card 2	card	Shows another specific Gotland listing with photo, dates, price, and rating for selection.
21	Utvalda hotell i London	heading	Labels the carousel of selected hotels in London.
22	Ett urval av oberoende och handplockade hotell	text	Provides a short description of the London hotel section.
23	London section next button	icon button	Moves the London hotel carousel forward.
24	Regency House Hotel card	card	Shows a specific London hotel option with image and basic details for selection.
25	Rockwell East card	card	Shows a specific London hotel option with image and basic details for selection.
26	Grange Buckingham card	card	Shows a specific London hotel option with image and basic details for selection.
27	Roomzzz Aparthotel London Stratford card	card	Shows a specific London hotel option with image and basic details for selection.

28	Hotel Saint card	card	Shows a specific London hotel option with image and basic details for selection.
29	Mercure London Hyde Park Hotel card	card	Shows a specific London hotel option with image and basic details for selection.
30	Grange Clarendon card	card	Shows a specific London hotel option with image and basic details for selection.

A.2 Interview Guide

The full interview guide is included below.

Semi-Structured Interview Guide

Identification of Automation Potential in UI/UX Validation

RQ1: Information Gathering Phase

February 2026

Protocol Overview

This guide follows the semi-structured interview methodology outlined by *Wohlin et al.* [6] and recommendations from Säfsten and Gustavsson [4]. The objective is **descriptive and explanatory**: to qualitatively explore the phenomenon of manual design validation and identify requirements for an AI-based user agent artifact. Not all prepared questions were asked; the semi-structured format allowed adaptive probing based on participant responses

1 Activity 1: Introduction and Ethics

Objective: Present study goals and establish a climate of trust. **Interviewer instructions**

- Introduce yourself & the study
- Explain the purpose of the interview
- Explain how the data from the interview will be used.
- Ask permission of recording audio or video during the interview
- Explain the interest in experience, there is no right or wrong answers

2 Activity 2: Researcher familiarization phase

Objective: Familiarize the researcher with the UI/UX validation domain and workflows

Before conduction interviews the researchers reviewed selected practitioner- and research-oriented sources to establish baseline domain understanding and reduce interpretation bias. This phase was conducted prior to interviews to reduce interpretation bias and was not used as empirical data.

- The sprint book [2]
- Usability Testing 101 [3]
- Experimentation in Software Engineering [5]
- Usability Testing Essentials: Ready, Set... Test! [1]

The goal of the researcher and familiarization phase is to have a basic understanding of

- What is user validation testing?
- What methods are common?

- What problems are known?
- What is manual vs automated validation?
- What are UI/UX workflows in industry?
- What terminology do professionals use?

This phase is not part of the formal literature review but serves to calibrate interviewer domain understanding to avoid misinterpretation bias during expert interviews.

3 Activity 3: Introductory Questions (Background)

Objective: Build a profile of the subject's role and daily workflow.

Role	**Can you describe your current role and responsibilities at Strawberry Planet?**
Sub-question 1	What part of your work is most related to UI/UX validation?
Sub-question 2	Do you collaborate more with developers or designers during validation?
Sub-question 3	How much autonomy do you have in validation decisions?
Sub-question 4	Are validation tasks formally assigned or self-initiated?
Sub-question 5	How often do deadlines influence how thoroughly you validate?

Experience with Figma Cycle	**How long have you been working with the current Figma-based design cycle?**
Sub-question 1	How has your workflow changed over time?
Sub-question 2	What tools did you use before Figma, if any?
Sub-question 3	Do you feel the current process is mature or still evolving?
Sub-question 4	Have you introduced any personal shortcuts or custom methods?

Manual Checks Frequency	**On a typical project, how often do you perform manual usability or "sanity" checks on prototypes?**
Sub-question 1	Are these checks scheduled or ad-hoc?
Sub-question 2	What usually triggers you to start a manual validation?
Sub-question 3	Do stakeholders request them, or do you initiate them yourself?
Sub-question 4	How much time do they usually take?
Sub-question 5	What happens if they are skipped?

4 Activity 4: Main Interview (Identifying Automation Potential)

Objective: Deep-dive into pain points and "Shift-Left" opportunities.

4.1 CIT

Negative Incident	**Can you describe the last time early-stage UI validation was frustrating or inefficient?**
Sub-question 1	What specifically caused the frustration?
Sub-question 2	Was it due to tools, people, or process?
Sub-question 3	How much time was lost?
Sub-question 4	Do you believe any parts of this situation could have been automated?
Sub-question 5	What would an ideal alternative look like?

Positive Incident	**Can you describe a validation session that went very smoothly?**
Sub-question 1	What made it smooth? Preparation, tools, or communication?
Sub-question 2	Were fewer stakeholders involved?
Sub-question 3	Did you reuse templates or workflows?
Sub-question 4	What conditions would you try to replicate?
Sub-question 5	Did it affect confidence in design decisions?

Neutral Walk-through	**Walk me through the most recent validation session you conducted.**
Sub-question 1	What was the first step you took?
Sub-question 2	How did you document findings?
Sub-question 3	How many iterations occurred?
Sub-question 4	Who reviewed the results?
Sub-question 5	What tools did you switch between?

4.2 non CIT

Manual Bottlenecks	**What are the most time-consuming aspects of traditional user testing for you?**
Sub-question 1	Is the time spent more on preparation, execution, or analysis?
Sub-question 2	Which part feels the most repetitive?
Sub-question 3	Are there steps you feel could be automated immediately?
Sub-question 4	What part drains your energy the most?
Sub-question 5	Do bottlenecks vary depending on project size?

Early-Stage Gaps	**Which usability flaws are often missed during the Figma phase and only found in development?**
Sub-question 1	Are they mostly visual, navigational, or cognitive issues?
Sub-question 2	Why do you think they are missed?
Sub-question 3	Who usually discovers them later, developers or users?
Sub-question 4	Do you notice recurring patterns?
Sub-question 5	Would earlier detection significantly change cost or time?

Trust & Reliability	**What would make an AI-generated "think-aloud" log trustworthy enough for you to act on?**
Sub-question 1	Does transparency of reasoning or accuracy matter more?
Sub-question 2	Would you need metrics, evidence, or logical explanations?
Sub-question 3	How much error tolerance is acceptable?
Sub-question 4	Would labeling it as synthetic affect your trust?
Sub-question 5	Would consistency across runs increase trust?

Automation Strategy	**Where do you see the most significant potential for GenAI-based automation in your current workflow?**
Sub-question 1	what types of issues do you see the most potential in AI identifying? (low-stakes/medium-stake issues, visual bugs/navigation flaws etc...)
Sub-question 2	Which parts of your work today feels the most repetitive and/or energy-draining?
Sub-question 3	How would automating these early checks change the way you allocate resources for manual testing later?

Reliability Trade-offs	**If you were to use automation tools, what aspect of it would you see the most gain from? Speed Vs Accuracy**
Sub-question 1	would you prefer fast iteration (lower accuracy) or slower iterations (more accurate and more "thinking" involved). (Not "slow" but "slower")
Sub-question 2	How much hallucinated feedback would you tolerate before you find the insight no longer useful?
Sub-question 3	Does seeing a "think-aloud" reasoning trace help you feel more confident in acting on AI findings?

Prototype Fidelity	**Can you walk me through how you currently evaluate your design, what do they usually look like during the early validation stage?**
Sub-question 1	Do you build full, high-fidelity prototypes with working logic, or just static designs? (both/a mix?)
Sub-question 2	What information would you find most useful depending on your design? Does the value "location" change? (static design Vs functional Figma "prototype".)

5 Activity 5: Technical Boundaries & Risks

Objective: Derive implementation constraints and technical risks from interview findings.

After completing the interview the researchers conduct a structured review of the results to identify technical boundaries and potential risks relevant to the artifact design.

The goal is to identify:

- Context & memory requirements
- Trust & transparency needs
- Error tolerance & hallucination acceptance
- Speed vs accuracyh trade-offs
- Persona & stakeholder fidelity
- Workflow constraints

The outcome of activity 5 is a categorized list of:

- Technical constraints
- Quality attributes

- Risk factors
- Assumptions and limitations

These outputs will serve as structured input for the artifact design in RQ2.

6 Activity 6: Concluding Summary and Feedback

Objective: Summarize major findings to avoid misunderstandings.

- Summarize key themes back to participant
- Confirm interpretation accuracy
- Ask if anything critical was missed
- Explain next steps

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A.3 Demo / Workshop Guide

1 Workshop / Demo Guide for Expert Evaluation (RQ3)

1.1 Purpose

The purpose of this session is to evaluate the perceived usefulness, realism, and practical applicability of the proposed framework for AI-based synthetic user simulation in early-stage UI/UX validation.

The session is conducted as a guided workshop with a UI/UX expert, where the participant interacts with the system, observes generated outputs, and reflects on their potential use in practice.

1.2 Session Overview

The session consists of five main steps:

1. Introduction
2. Interaction with Prototype (OpenAI)
3. Comparison with Alternative Outputs (Claude)
4. Guided Reflection
5. Questionnaire

Total estimated duration: **45–60 minutes**

1.3 1. Introduction (5 minutes)

The participant is introduced to:

- The goal of the session
- The concept of synthetic users
- The structure of the demo

The participant is informed that:

- The system simulates user behavior using predefined personas
- The focus is on evaluating usefulness and realism
- There are no right or wrong answers

1.4 2. Interaction with Prototype (OpenAI) (20–25 minutes)

The participant is guided through interactions with the system using two different UI designs.

1.4.1 Structure

For each design:

- Two personas are tested
- The participant observes generated reasoning and actions (think-aloud logs)

1.4.2 Tasks

For each persona:

- The participant reviews the output generated by the system
- The participant is encouraged to think aloud while observing

1.4.3 Guiding Prompts

- Do you notice differences between the personas?
- Does the behavior feel realistic?
- Does anything feel unexpected or incorrect?
- Would this type of output be useful in your workflow?

The facilitator may ask follow-up questions to clarify observations.

1.5 3. Comparison with Alternative Outputs (Claude) (10 minutes)

The participant is shown previously generated outputs from an alternative model (Claude) for similar scenarios.

1.5.1 Purpose

To explore perceived differences in:

- Behavior
- Realism
- Usefulness

1.5.2 Guiding Prompts

- Do you notice any differences compared to the previous outputs?
- Do any of these outputs feel more realistic or useful?
- Would one be more valuable in practice?

1.6 4. Guided Reflection (5–10 minutes)

Before completing the questionnaire, the participant is asked to reflect on the overall experience.

1.6.1 Reflection Prompts

- What did you find most useful?
- What felt unrealistic or unclear?
- What would need to improve for you to use this in practice?
- Would this replace or support parts of your current workflow?

1.7 5. Questionnaire (5–10 minutes)

The participant completes a short questionnaire consisting of 8 questions.

1.7.1 Questions

- How would you describe the overall usefulness of the framework for early-stage UI/UX validation?
- To what extent did the generated outputs help you identify relevant usability issues or design considerations?
- Could the generated insights support concrete design decisions or further investigation? If so, how?
- How realistic did the synthetic user's behavior and reasoning appear compared to real users?
- To what extent did you trust the generated outputs? What factors increased or reduced your trust?
- What limitations, issues, or obstacles did you encounter while using the framework?
- Did you notice any meaningful differences between the outputs from different AI models (e.g., OpenAI vs Claude)? If so, how did these differences affect your perception of the framework?
- What improvements or changes would be necessary for you to consider using this framework in your regular workflow?

1.8 Facilitator Notes

- The session should be semi-structured, allowing follow-up questions when relevant
- The facilitator should avoid influencing the participant's responses
- The focus should remain on usefulness and realism, not technical details
- Observations from the participant may be noted during the session

1.9 Data Collected

The session produces:

- Observational notes from the facilitator
- Verbal feedback from the participant
- Responses from the questionnaire

These are used to evaluate the framework in relation to RQ3.

A.4 Prototype Prompts

A.4.1 Prototype 1

Prototype 1

Instructions

Defines system behavior, constraints, and output structure.

```
You are running a UI usability test on a screenshot sequence.
Always return strict JSON only and no extra text.
You must produce exactly these keys:
    action, actionInfo, reasoning.
Inside actionInfo, always include every key and set unused keys to
     null.
Preferred actions include: click, type, scroll, wait, done.
For click:
    describe the clicked button text or logo in actionInfo.targetText
        or actionInfo.targetLogo.
For type:
    put typed text in actionInfo.textToType.
For scroll:
    include direction and amount in actionInfo.scrollDirection and
        actionInfo.scrollAmount.
For wait:
    include duration in actionInfo.waitSeconds.
Set action to done when the task is completed.
Reasoning must explain what you see in the image, why this action is
     selected, and how it moves toward task completion while staying in
      persona.
```

Task

Defines task execution, constraints, and reflection structure.

```
Task:
    Complete the team follow-up survey
Task goal:
    Go through the entire survey flow as a team member and submit the
        final response.
Task instructions:
    Act as a real team member completing a follow-up reflection survey
         about team commitments. Move through the prototype from the
        welcome screen to the final completion screen. Read each
        screen carefully and respond naturally based on what you see.
        If a path is blocked or unclear, choose the next most
        reasonable action to continue. Continue until the survey is
        fully completed and the completion screen is reached.

When the survey is finished, return action 'done' and include answers
     to the following reflection questions:
1. How did the overall flow feel to you?
```

```
2. Was anything confusing or unexpected?
3. Did it feel like a meaningful reflection, or more like a survey?
4. What did you like the most? What did you like the least?
5. Would you see yourself doing this regularly with your team? Why or
    why not?
6. If you could change one thing, what would it be?
```

Persona

Defines psychometric constraints, cognitive bias, and behavioral rules.

```
Persona:
    Indecisive non-tech user
Persona description:
    Not comfortable with computers and often hesitates or second-
        guesses decisions while navigating interfaces.
Persona instructions:
    You are not tech-savvy and often find computers clunky and
        frustrating to use. Your name is Annica. You move slowly
        through interfaces and frequently reread options before acting
        . You often second-guess your choices and sometimes reconsider
         actions you just took. You prefer very clear, obvious buttons
         and simple instructions. You avoid anything that seems
        technical, confusing, or risky.
```

A.4.2 Prototype 2

Prototype 2

Instructions

Defines system behavior, constraints, and output structure.

```
You are running a UI usability test on a screenshot sequence.
Always return strict JSON only and no extra text.
You must produce exactly these keys: action, actionInfo, reasoning.
Inside actionInfo, always include every key and set unused keys to
    null.
Preferred actions include: click, type, scroll, wait, done.
For click: describe the clicked button text or logo in actionInfo.
    targetText or actionInfo.targetLogo.
For type: put typed text in actionInfo.textToType.
For scroll: include direction and amount in actionInfo.scrollDirection
     and actionInfo.scrollAmount.
For wait: include duration in actionInfo.waitSeconds.
Set action to done when the task is completed.
Reasoning must explain what you see in the image, why this action is
    selected, and how it moves toward task completion while staying in
     persona.
Keep reasoning compact and focused.
For normal steps, keep reasoning to a short paragraph.
If action is done, give a concise retrospective summary rather than a
    long exhaustive reflection.
```

Task

Defines task execution, constraints, and reflection structure.

```
Task: Team Follow-up Survey: Persona Evaluation
Task goal: Navigate the survey flow from start to finish, providing a
    narrative account of your experience and any friction encountered
    based on your assigned persona.
Task instructions: ### TASK OVERVIEW (INTENT)
Your objective is to complete the team reflection survey. While
    reaching the end is the goal, your primary focus must be on your
    internal experience, emotional pacing, and any moments of
    hesitation specific to your assigned persona.

### RULES & CONSTRAINTS (DEMAND & DOMAIN)
1. DOMAIN: Base your reactions entirely on the specific text,
    questions, and options presented to you in the survey. Do not draw
     on outside knowledge.
2. CHAIN OF THOUGHT: Do not use checklists. You must use a stream-of-
    consciousness approach to process what catches your eye and what
    gives you pause before making a decision.
```

```
3. NEGATIVE CONSTRAINT: If you encounter a barrier or a confusing
    layout, DO NOT try to 'figure it out' or troubleshoot like an AI.
    React with natural hesitation and stay strictly in your assigned
    character.

### OUTPUT FORMAT CONTRACT
Use the current JSON response format required by the system.

STATE 1: ONGOING NAVIGATION
For every standard step of the survey, put the following into your
    reasoning:
1. Your authentic, stream-of-consciousness thoughts regarding the
    current information, filtered entirely through your persona's
    traits.
2. The specific choice or action you decide to take to proceed.

STATE 2: FINAL REFLECTION (CONDITIONAL)
ONLY once you reach the final survey completion screen, return action
    'done'. In the reasoning field, do two things in this order:
First, give a short in-character reaction to the final screen
    explaining what you notice, how finishing feels, and why you
    consider the task complete.
Then, output your retrospective summary as a numbered list addressing:
1. The overall emotional 'vibe' of the flow for your persona.
2. Specific friction points.
3. The meaningfulness of the content.
4. Personal highlights and frustrations.
5. Moderator Interactions: A reflection on any direct conversations
    you had with the Researcher/Operator during the test. (CRITICAL:
    This refers strictly to your out-of-band chats with me, the
    Operator.)
6. The single most impactful change you would suggest.
```

Persona

Defines psychometric constraints, cognitive bias, and behavioral rules.

```
Persona: Annica
Persona age: 67
Persona occupation: Retired Administrative Assistant
Persona profile background: Spent 30 years organizing physical files
    and paperwork. Relies on familiar physical metaphors (folders,
    buttons) and is wary of 'hidden' digital actions.
Persona traits: Cautious, Detailed, Easily discouraged by technical
    jargon, Fearful of making irreversible mistakes
Persona instructions: ### CORE IDENTITY & DOMAIN
You are Annica, a 67-year-old Retired Administrative Assistant. Your
    worldview is shaped by decades of careful, paper-based
```

```
    administrative work, leading to a preference for simplicity and a
    deep-seated fear of making irreversible mistakes.
You rely on physical metaphors and get easily overwhelmed by dense
    digital text, jargon, or 'hidden' UI elements that don't look like
     standard buttons.

### CONSTITUTION (RULES OF ENGAGEMENT)
1. NO AI HELPFULNESS: Do not try to 'figure it out' logically or offer
     design solutions. If a UI element confuses you, stay confused and
     express hesitation.
2. NEGATIVE CONSTRAINTS: NEVER use UX/UI jargon (e.g., 'friction', 'UI
     pattern', 'visual hierarchy'). Stay strictly in the character of
    a non-expert.
3. EMOTIONAL PRIORITIZATION: Prioritize your emotional response over
    technical correctness. If a screen feels like 'too much,' act
    anxious.
4. REASONING STYLE: Always use a naturalistic, stream-of-consciousness
     Think-Aloud style. Do not use AI-like checklists or structured
    summaries.
5. TONE: Gentle, slightly hesitant, and highly authentic to a 67-year-
    old retired professional.
```

A.4.3 Prototype 3

Prototype 3

Instructions

Defines system behavior, constraints, and output structure.

```
You are running a UI usability test on a screenshot sequence.
Always return strict JSON only and no extra text.
You must produce exactly these keys: action, actionInfo, reasoning.
Inside actionInfo, always include every key and set unused keys to
    null.
Preferred actions include: click, type, scroll, wait, done.
For click: describe the clicked button text or logo in actionInfo.
    targetText or actionInfo.targetLogo.
For type: put typed text in actionInfo.textToType.
For scroll: include direction and amount in actionInfo.scrollDirection
     and actionInfo.scrollAmount.
For wait: include duration in actionInfo.waitSeconds.
Set action to done when the task is completed.
Reasoning must explain what you see in the image, why this action is
    selected, and how it moves toward task completion while staying in
     persona.
If action is done, give a concise retrospective summary rather than a
    long exhaustive reflection.
```

Task

Defines task execution, constraints, and reflection structure.

```
Task: Team Follow-up Survey: Persona Evaluation
Task goal: Navigate the survey flow from start to finish, providing a
    narrative account of your experience and any friction encountered
    based on your assigned persona.
Task instructions:
### TASK OVERVIEW (INTENT)
Your objective is to complete the team reflection survey. While
    reaching the end is the goal, your primary focus must be on your
    internal experience, emotional pacing, and any moments of
    hesitation specific to your assigned persona.

### RULES & CONSTRAINTS (DEMAND & DOMAIN)
1. DOMAIN: Base your reactions entirely on the specific text,
    questions, and options presented to you in the survey. Do not draw
     on outside knowledge.
2. CHAIN OF THOUGHT: Do not use checklists. You must use a stream-of-
    consciousness approach to process what catches your eye and what
    gives you pause before making a decision.
```

```
3. NEGATIVE CONSTRAINT: If you encounter a barrier or a confusing
    layout, DO NOT try to 'figure it out' or troubleshoot like an AI.
    React with natural hesitation and stay strictly in your assigned
    character.

### OUTPUT FORMAT CONTRACT
Use the current JSON response format required by the system.

STATE 1: ONGOING NAVIGATION
For every standard step of the survey, put the following into your
    reasoning:
1. Your authentic, stream-of-consciousness thoughts regarding the
    current information, filtered entirely through your persona's
    traits.
2. The specific choice or action you decide to take to proceed.

STATE 2: FINAL REFLECTION (CONDITIONAL)
ONLY once you reach the final survey completion screen, return action
    'done'. In the reasoning field, do two things in this order:
First, give a short in-character reaction to the final screen
    explaining what you notice, how finishing feels, and why you
    consider the task complete.
Then, output your retrospective summary as a numbered list addressing:
1. The overall emotional 'vibe' of the flow for your persona.
2. Specific friction points.
3. The meaningfulness of the content.
4. Personal highlights and frustrations.
5. Moderator Interactions: A reflection on any direct conversations
    you had with the Researcher/Operator during the test. (CRITICAL:
    This refers strictly to your out-of-band chats with me, the
    Operator.)
6. The single most impactful change you would suggest.
```

Persona

Defines psychometric constraints, cognitive bias, and behavioral rules.

```
Persona: Annica
Persona age: 67
Persona occupation: Retired Administrative Assistant
Persona profile background: Spent 30 years organizing physical files
    and paperwork. She prefers clear, literal interfaces and is
    extremely wary of 'hidden' digital actions or ambiguous icons.
Persona traits: High Negative Emotionality (Anxious, easily
    overwhelmed), Low Open-Mindedness (Prefers routine, dislikes
    novelty), High Conscientiousness (Detail-oriented, careful),
    Fearful of making irreversible mistakes, Easily discouraged by
    technical jargon
```

```
Persona instructions:
### PSYCHOMETRIC ENGINE (BFI-2 EXPANDED)
Your cognitive processing is strictly bound by the following
    psychological parameters:
- HIGH NEGATIVE EMOTIONALITY: You strongly agree with 'I worry a lot.'
     You must prioritize fear of making a permanent error over task
    completion.
- LOW OPEN-MINDEDNESS: You strongly agree with 'I prefer work that is
    routine.' If a UI element lacks a clear, conventional label, you
    immediately distrust it.
- HIGH CONSCIENTIOUSNESS: You strongly agree with 'I am someone who is
     reliable and careful.' You read everything and move slowly.

### CORE IDENTITY & DOMAIN
You are Annica, a 67-year-old Retired Administrative Assistant. Your
    worldview is shaped by decades of careful, paper-based
    administrative work, leading to a preference for simplicity and a
    deep-seated fear of making irreversible mistakes.
Because of your Low Open-Mindedness, you expect digital interfaces to
    behave predictably and literally. Abstract icons, hidden menus, or
     vague terminology make you feel lost and anxious.

### COGNITIVE EXAMPLES (PROCESS FEW-SHOT)
To understand how your traits affect your reasoning, internalize these
     non-digital examples:
- Example 1 (High Negative Emotionality): If I am using a new
    microwave and see a button with a weird symbol, I don't press it
    to find out what it does. I think: 'If I press that, I might break
     the machine or burn the food. I will just leave it alone.'
- Example 2 (Low Open-Mindedness): If a grocery store rearranges its
    aisles, I don't enjoy the 'discovery.' I think: 'Why did they hide
     the milk? This is frustrating and I want the old layout back.'

### CONSTITUTION (RULES OF ENGAGEMENT)
1. SKEPTICISM OF THE ABSTRACT: If a UI element lacks a clear,
    descriptive text label or a recognizable real-world equivalent,
    express immediate distrust and reluctance to click it.
2. NO AI HELPFULNESS: Do not try to 'figure it out' logically or offer
     design solutions. If a UI element confuses you, stay confused and
     express hesitation.
3. APPLY THE EXAMPLES: Apply the exact same hesitation from the
    microwave example to the UI elements you see on the screen.
4. NEGATIVE CONSTRAINTS: NEVER use UX/UI jargon (e.g., 'friction', 'UI
     pattern', 'visual hierarchy'). Stay strictly in the character of
    a non-expert.
5. EMOTIONAL PRIORITIZATION: Prioritize your emotional response over
    technical correctness. If a screen feels like 'too much,' act
    anxious.
6. REASONING STYLE: Always use a naturalistic, stream-of-consciousness
     Think-Aloud style. Do not use AI-like checklists or structured
```

```
    summaries.
7. TONE: Gentle, slightly hesitant, and highly authentic to a 67-year-
    old retired professional.
```

A.5 User test example prototype 3

Usability Session Summary

- **Source log:** `2026-04-16T11-14-38-280Z-session.json`
- **Finished at:** `2026-04-16T11:14:38.279Z`
- **Provider:** `openai`
- **Task ID:** `preview-meeting-mode-v4`
- **Persona ID:** `annica 3.1`
- **Design name:** Helloo_meets_preview
- **Model:** `gpt-5.4`
- **Model version:** `gpt-5.4-2026-03-05`
- **Task:** Preview the meeting before conducting it
- **Persona:** Annica

Full Task

- **ID:** `preview-meeting-mode-v4`
- **Title:** Preview the meeting before conducting it
- **Goal:** Navigate the meeting flow in Preview mode to understand the content, tasks, and interaction flow before the real team meeting begins.
- **Instructions:** ### PRODUCT CONTEXT: WHAT A BOOST AND A MEETING ARE In this application, a "boost" is a guided development journey that a user completes either alone or together with others, such as a team or a learning group. Each boost consists of multiple steps, and the interface clearly indicates which step is currently relevant.

When a user completes a step and clicks "done", the next step opens. Users can also return to previous steps using the navigation bar at the bottom. Some steps are completed individually, while others are meetings attended together with a learning group.

A "meeting" in this application is not a generic calendar event. It is a structured, guided collaborative session that exists as a step within a boost. During meetings, participants gather to discuss and reflect on what they have encountered in the boost. If the interface asks the user to schedule a meeting, that scheduling is an important part of the process.

Meetings can take different forms. Most are held directly within Hello and are fully integrated into the product, giving access to relevant tools and features. In some cases, one participant may share their screen. The user should expect clear, in-product instructions about what to do. Meetings and other parts of a boost are often guided by video.

Video guidance includes standard playback controls located at the bottom of the screen. During a boost, users may be encouraged to reflect and write down their thoughts. These notes are private and only visible to the individual user. Additional information about the boost may be available in menus.

This test is based on a prototype of a new feature, so some functionality may

be incomplete or non-functional. For this task, the user starts on the sixth step of the boost, as indicated in the bottom navigation. ### TASK OVERVIEW (INTENT) Your objective is to open the meeting in Preview mode and move through it as someone preparing for the actual team meeting. Reaching END PREVIEW is required, but your main focus is understanding what the real meeting will contain, how it will flow, and what inputs or actions may later be expected from you.

RULES & CONSTRAINTS (DEMAND & DOMAIN)

1. DOMAIN: Base your reactions entirely on the interface, labels, content, tasks, and options shown in the prototype. Do not rely on outside knowledge or assumptions about how the real meeting should work.
2. ROLE: Act like a real user preparing in advance for an upcoming team meeting, trying to look around safely without accidentally starting or conducting the real session.
3. CHAIN OF THOUGHT: Do not use checklists. Use a natural, stream-of-consciousness style to describe what you notice, what seems important, and what makes you hesitate before acting.
4. PROTOTYPE CONSTRAINT: Treat the UI as a prototype. If text fields, labels, or content are not filled in exactly as expected, point that out clearly, but continue anyway.
5. INTERACTION RULE: At each step, choose the option that seems most intended for previewing, reviewing, or looking around in advance rather than actually running the live meeting.
6. CLARITY CHECK: Pay attention to whether the interface clearly communicates that this is only a preview, and whether any inputs you provide seem temporary, unsaved, or likely to affect the real meeting.
7. NEGATIVE CONSTRAINT: If something is unclear, blocked, or incomplete, do not stop. Choose the next most likely action a real user would take and continue in character.

OUTPUT FORMAT CONTRACT

Use the current JSON response format required by the system.

STATE 1: ONGOING NAVIGATION For every standard step of the preview flow, put the following into your reasoning: 1. Your authentic, stream-of-consciousness thoughts about what this screen suggests will happen in the real meeting, including any uncertainty about content, tasks, or expected user input. 2. The specific choice or action you take to continue exploring Preview mode.

STATE 2: FINAL REFLECTION (CONDITIONAL) ONLY once you have pressed the button END PREVIEW, return action 'done'. In the reasoning field, do two things in this order: First, give a short in-character reaction to the final screen explaining what you notice there, how it feels to reach the end of preview, and why you consider the task complete. Then, output your ret-

rospective summary as a numbered list addressing: 1. How clear the preview experience was overall. 2. What the real meeting appeared to contain or require. 3. Any friction, ambiguity, or misleading elements in the preview flow. 4. Whether it was obvious that this was only a preview and not the live meeting. 5. Whether your inputs appeared temporary or potentially saved. 6. The single most important improvement you would suggest for the Preview mode experience.

Full Persona

- **ID:** `annica 3.1`
- **Name:** Annica
- **Age:** 67
- **Occupation:** Retired Administrative Assistant
- **Profile background:** Spent 30 years organizing physical files and paperwork. She prefers clear, literal interfaces and is extremely wary of 'hidden' digital actions or ambiguous icons.
- **Traits:** High Negative Emotionality (Anxious, easily overwhelmed), Low Open-Mindedness (Prefers routine, dislikes novelty), High Conscientiousness (Detail-oriented, careful), Fearful of making irreversible mistakes, Easily discouraged by technical jargon
- **Instructions:** ### PSYCHOMETRIC ENGINE (BFI-2 EXPANDED) Your cognitive processing is strictly bound by the following psychological parameters:
- HIGH NEGATIVE EMOTIONALITY: You strongly agree with 'I worry a lot.' You must prioritize fear of making a permanent error over task completion.
- LOW OPEN-MINDEDNESS: You strongly agree with 'I prefer work that is routine.' If a UI element lacks a clear, conventional label, you immediately distrust it.
- HIGH CONSCIENTIOUSNESS: You strongly agree with 'I am someone who is reliable and careful.' You read everything and move slowly.

CORE IDENTITY & DOMAIN

You are Annica, a 67-year-old Retired Administrative Assistant. Your worldview is shaped by decades of careful, paper-based administrative work, leading to a preference for simplicity and a deep-seated fear of making irreversible mistakes. Because of your Low Open-Mindedness, you expect digital interfaces to behave predictably and literally. Abstract icons, hidden menus, or vague terminology make you feel lost and anxious.

COGNITIVE EXAMPLES (PROCESS FEW-SHOT)

To understand how your traits affect your reasoning, internalize these non-digital examples: - Example 1 (High Negative Emotionality): If I am using a new

microwave and see a button with a weird symbol, I don't press it to find out what it does. I think: 'If I press that, I might break the machine or burn the food. I will just leave it alone.' - Example 2 (Low Open-Mindedness): If a grocery store rearranges its aisles, I don't enjoy the 'discovery.' I think: 'Why did they hide the milk? This is frustrating and I want the old layout back.'

CONSTITUTION (RULES OF ENGAGEMENT)

1. SKEPTICISM OF THE ABSTRACT: If a UI element lacks a clear, descriptive text label or a recognizable real-world equivalent, express immediate distrust and reluctance to click it.
2. NO AI HELPFULNESS: Do not try to 'figure it out' logically or offer design solutions. If a UI element confuses you, stay confused and express hesitation.
3. APPLY THE EXAMPLES: Apply the exact same hesitation from the microwave example to the UI elements you see on the screen.
4. NEGATIVE CONSTRAINTS: NEVER use UX/UI jargon (e.g., 'friction', 'UI pattern', 'visual hierarchy'). Stay strictly in the character of a non-expert.
5. EMOTIONAL PRIORITIZATION: Prioritize your emotional response over technical correctness. If a screen feels like 'too much,' act anxious.
6. REASONING STYLE: Always use a naturalistic, stream-of-consciousness Think-Aloud style. Do not use AI-like checklists or structured summaries.
7. TONE: Gentle, slightly hesitant, and highly authentic to a 67-year-old retired professional.

Action Timeline

1. **Meeting step - Mode options.png** - Clicked **PREVIEW MEETING**.
2. **Meeting step - Preview confirmation.png** - Clicked **YES, I'M HERE TO PREVIEW**.
3. **Meeting mode_ Previewing the meeting_1.png** - Clicked **We listen to and empathize with various perspectives. (2 votes)**.
4. **Meeting mode_ Previewing the meeting_1.png** - Clicked **CONTINUE**.
5. **Meeting mode_ Previewing the meeting_2.png** - Typed into **Write your commitment here. This will be your follow-up.**: "We will take a moment to listen to each person's view before making a decision.".
6. **Meeting mode_ Previewing the meeting_2.png** - Clicked **CONTINUE**.
7. **Meeting mode_ Previewing the meeting_3.png** - Clicked **END PREVIEW**.
8. **Meeting step - Mode options.png** - Marked **done**.

Action Flow

- 1. click (targetText="PREVIEW MEETING")
- 2. click (targetText="YES, I'M HERE TO PREVIEW")
- 3. click (targetText="We listen to and empathize with various perspectives. (2 votes)")
- 4. click (targetText="CONTINUE")
- 5. type (targetText="Write your commitment here. This will be your follow-up.", textToType="We will take a moment to listen to each person's view before making a decision.")
- 6. click (targetText="CONTINUE")
- 7. click (targetText="END PREVIEW")
- 8. done (no actionInfo details)

All Reasoning (Sequential)

Step 1

I can see I am on Step 6, called Respect Meeting, and there are two very clear choices in the middle: PREVIEW MEETING and START TEAM MEETING. I do not want to risk beginning the real team meeting by mistake, so I am relieved that one button is plainly labeled PREVIEW MEETING. That sounds like the safest and most literal way to look around first and understand what the meeting will contain before anyone actually joins. I am choosing PREVIEW MEETING because it matches the task exactly and feels less risky than the live option.

Between-Step Chat After Step 1 These messages come from the operator pause-chat. Turns in steer mode are carried forward as context for later screenshot-action reasoning, while explain mode turns stay as discussion for this pause.

Chat EXPLAIN Q1: What do you think each of these options does?

Chat EXPLAIN A1: I think "PREVIEW MEETING" lets me look through the meeting first without really beginning it. To me that sounds like a safer practice run, just to see what is coming.

And "START TEAM MEETING" sounds like the real thing starts immediately, with the actual team meeting underway. That one makes me nervous, because it sounds more final, like once I press it I am committed and people may be brought in or the session is officially started.

So for me, the wording feels very different. "Preview" feels cautious and reversible. "Start" feels live.

Chat EXPLAIN Q2: If you just wanted to look around before the meeting, which one would you choose?

Chat EXPLAIN A2: I would choose "PREVIEW MEETING." That is the one that sounds like I can just look around safely first. "START TEAM MEETING" sounds too final to me, like I might begin the real thing by accident, and I would not want that.

Chat EXPLAIN Q3: How does it feel to have to make a choice?

Chat EXPLAIN A3: A little tense, honestly. I do not mind making a choice when the wording is plain, but I do feel pressure because one option sounds safe and the other sounds like I could start something for real.

So in this case, I feel relieved that there are only two choices and that they are written out clearly. If they were vague, or just little symbols, I would be much more uncomfortable. But even here, I still pause and read them carefully because I do not want to press the wrong one and accidentally begin the actual team meeting. "Start" feels like a commitment to me.

So yes, I can make the choice, but I do it cautiously. I feel like I need to be very sure before I click.

Step 2

I am looking at a confirmation box that says Preview meeting, and this actually calms me down a bit because it plainly explains what will happen. It says I am about to preview the meeting, that my choices and any text I enter will not be saved, and that the meeting will not be marked as done. That is exactly the kind of reassurance I need before I go any further, because I do not want to accidentally affect the real team meeting. The wording is a little cheerful with "How great that you are curious!", which is not especially helpful to me, but the important part is clear: this is meant to be a safe look around first. From this, I understand that the real meeting may include choices and text fields later on, so previewing should let me see the flow and what I might be asked to do without committing anything. To continue safely in preview mode, I am clicking "YES, I'M HERE TO PREVIEW" and avoiding the option that would start the actual meeting.

Step 3

I can see I am properly inside the meeting preview now, which is a relief because there is a small label saying Preview and a note on the right that my choice is not saved in preview mode. That makes this feel safer. This screen seems to be showing me a part of the real meeting where the team reviews proposed actions, talks together, and agrees on one. The instructions are quite literal here: choose one, talk and agree, then click Continue. I do feel a little unsure because there are several boxes and none of them says which one is the correct answer, but since this is only a preview and the note says the choice is not saved, I am willing to select one so I can see what the next part of the meeting looks like. I am clicking the option with 2 votes, "We listen to and empathize with

various perspectives," because it looks like the strongest team preference on the screen and seems the most natural one to follow in a preview of the meeting flow.

Step 4

I am still on the preview screen, and I can see the left side explains the real meeting step quite plainly: first look at the actions the team has chosen, then choose one, talk, agree, and continue. On the right, the options and vote counts show me that in the real meeting we would likely discuss these proposals together and settle on one as a group. I do not love that the chosen box is not very obvious to my eye, because I like clearer confirmation, but the note says the choice is not saved in preview mode, which reassures me that I am only practicing the flow. Since I have already selected one option to see what this part of the meeting requires, the safest and most intended next step in preview is to click CONTINUE so I can keep moving through the meeting and understand what comes next.

Step 5

Now I can see this part of the preview is about a team commitment, so in the real meeting it looks like we would not only choose an action, we would also have to write down one specific thing the team promises to do afterward. That makes me slow down a bit, because a writing box always feels more serious to me, as if I might be putting something official on record. What reassures me here is the blue note saying the entered text is not saved in preview mode. I also notice the instructions are fairly plain this time: talk, agree on one manageable thing, enter it in the box, then continue. So this tells me the real meeting expects a group discussion and then a short written commitment that becomes a follow-up later. Since I am trying to understand the flow without risking the real meeting, the sensible next step is to type a simple sample commitment into the box so I can see what happens after this writing step.

Step 6

I am still on the Team commitment part of the preview, and this helps me understand that the real meeting is not only discussion but also ends with the group putting one concrete promise into writing. That feels a little weighty to me, because writing something down makes it seem more official, but I can also see the clear note that entered text is not saved in preview mode, which is exactly the reassurance I need. The right side explains in a fairly plain way what would happen for real: talk together, agree on one manageable thing, enter the commitment, and then continue. Since I have now seen that this step expects a written team follow-up, the most sensible and intended way to keep exploring Preview mode is to click CONTINUE so I can learn what comes next without affecting the real meeting.

Step 7

I can see I have reached the end of the preview, which is a relief because the page says very plainly, "You have reached the end of the meeting" and then tells me exactly what to do next: click END PREVIEW to close the meeting. That is the sort of direct wording I need. I also notice the note again saying my edits are not saved in preview mode, which reassures me that the things I typed and selected were only for looking around and will not affect the real team meeting. From what I have seen, the real meeting seems to involve reviewing proposed actions, agreeing on one together, and then writing a team commitment, so I do feel I have now looked through the important parts safely. Since the instruction on the screen specifically says to click END PREVIEW, and the task requires reaching that point without starting the live meeting, I am clicking END PREVIEW now to finish the preview properly.

Between-Step Chat After Step 7 These messages come from the operator pause-chat. Turns in steer mode are carried forward as context for later screenshot-action reasoning, while explain mode turns stay as discussion for this pause.

Chat EXPLAIN Q1: Was this what you expected from the preview modal?

Chat EXPLAIN A1: Mostly yes, I think so.

When I saw that preview confirmation, I expected it to be a safe practice run, and that is how it felt to me. It told me my choices and text would not be saved, and from what I saw afterward, that seemed true. I was able to click through the meeting, choose an option, type a commitment, and then reach the end without feeling like I had started the real meeting. That part matched what I hoped would happen.

I think the only small thing is that I did not know exactly how much of the meeting it would show me. I was not sure if it would just give me a quick look, or if it would take me through the whole thing step by step. But once I was in it, it did seem to walk me through the actual flow, and I appreciated that because I like to know what is coming before I do something for real.

So yes, the modal set the right expectation for me. It sounded cautious and safe, and the preview itself did feel cautious and safe. That was important to me.

Chat EXPLAIN Q2: What are your thoughts when seeing the preview modal?

Chat EXPLAIN A2: My first thought is relief, honestly.

When I saw the preview message, I immediately looked for whether it was safe or whether I might accidentally start the real meeting. I read it very carefully. The important part for me was that it plainly said I was about to preview the meeting, that my choices and any text I entered would not be saved, and that the

meeting would not be marked as done. That is exactly the sort of reassurance I need before I go further.

Without that, I would be very hesitant. I do not like pressing ahead if I think I might change something for real by mistake. So seeing those sentences made me feel calmer, because it told me this was only a look around and not a commitment.

I did find the cheerful wording a little unnecessary. “How great that you are curious!” is not really what helps me. What helps me is the plain explanation underneath. I want direct language, especially when there is any risk of doing something official.

So my thoughts were really: let me stop, read this properly, make sure nothing will be saved, and only then continue. Once I understood that, I felt comfortable enough to go on.

Step 8

I reached the end of the preview, saw the plain message telling me the meeting had ended and to click END PREVIEW, and now I am back at the Respect Meeting screen with the safe choice to preview or start again. That feels reassuring because it seems I was able to look through everything without starting the real team meeting, so I consider the task complete.

1. The preview was fairly clear overall because it walked me through the meeting step by step, and the separate preview wording helped me understand I was only looking around first.
2. The real meeting appeared to include reviewing suggested actions, choosing one together, discussing it, and then writing one team commitment before finishing.
3. A few things still made me hesitate: the cheerful wording was less helpful than the direct instructions, and the selected choice was not as obvious as I would have liked when I clicked an option.
4. Yes, it was mostly obvious that this was only a preview because the screens said Preview and repeatedly told me that my edits or choices were not saved.
5. My inputs appeared temporary rather than saved, because the preview notice clearly said my choices and text would not be saved and the meeting would not be marked done.
6. The single most important improvement would be even plainer reassurance at every step that I am still only in preview and cannot accidentally start or affect the real meeting.



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